

Middle East Technical University

Department of Electrical and Electronics Engineering

EE568 – Design of Electrical Machines

Project 2: Motor Winding Design & Analysis

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Design: Design 1: $N_s = 15$, $N_m = 8$

Repository: github.com/seferidaci/EE568-Projects-sefer_idaci

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1 Q1 – Integral-Slot Winding Design

1.1 Machine Parameters and Notation

The given machine has the following specifications:

Table 1: Given and derived parameters – Q1

Parameter	Meaning	Symbol	Value
Number of stator slots	Total slots around the stator bore	N_s	72
Number of poles	Total magnetic poles on the rotor	$2p$	6
Number of pole pairs	Half the pole count; $p = 2p/2$	p	3
Number of phases	Phases of the AC supply	m	3
Winding layers	Number of coil sides per slot	—	2 (double-layer)

1.2 Step 1 – Derived Quantities

1.2.1 Electrical Angle per Slot α_e

$$\boxed{\alpha_e = \frac{p \times 360^\circ}{N_s}} = \frac{3 \times 360^\circ}{72} = \mathbf{15^\circ} \text{ (electrical)} \quad (1)$$

1.2.2 Slots per Pole per Phase q

The total number of slots is shared equally among $2p$ poles and m phases:

$$\boxed{q = \frac{N_s}{2p \cdot m}} = \frac{72}{6 \times 3} = \mathbf{4} \quad (2)$$

Because $q = 4 \in \mathbb{Z}^+$, this is an *integral-slot* winding. Each phase occupies 4 consecutive slots under each pole.

1.2.3 Pole Pitch τ_p

The pole pitch is the number of slots under one pole:

$$\boxed{\tau_p = \frac{N_s}{2p}} = \frac{72}{6} = \mathbf{12} \text{ slots} \quad (3)$$

A *full-pitched* coil spans exactly $\tau_p = 12$ slots.

1.2.4 One Pole-Pair = 360° electrical = 24 slots

Since $\alpha_e = 15^\circ$, it takes $360^\circ/15^\circ = 24$ slots to complete one full electrical cycle. This means **one pole-pair spans 24 consecutive slots**, which is the minimum repeating unit of the winding.

1.3 Step 2 – Upper Layer Assignment (the 60° Sector Rule)

1.3.1 Physical Basis

Each stator slot has a conductor whose induced EMF is represented by a phasor. The phasor of slot k (upper layer) points at electrical angle:

$$\varphi_k = (k - 1) \alpha_e \pmod{360^\circ} \quad (4)$$

Sector	Angle range	Assignment
1	$[0^\circ, 60^\circ)$	A+
2	$[60^\circ, 120^\circ)$	C–
3	$[120^\circ, 180^\circ)$	B+
4	$[180^\circ, 240^\circ)$	A–
5	$[240^\circ, 300^\circ)$	C+
6	$[300^\circ, 360^\circ)$	B–

1.3.2 Slot Angle Table – One Pole-Pair (Slots 1–24)

Table 2: Upper-layer phase assignment for one pole-pair (72-slot, 6-pole)

Slot k	φ_k	Sector	Layer-1	Slot k	φ_k	Sector	Layer-1
1	0°	1	A+	13	180°	4	A–
2	15°	1	A+	14	195°	4	A–
3	30°	1	A+	15	210°	4	A–
4	45°	1	A+	16	225°	4	A–
5	60°	2	C–	17	240°	5	C+
6	75°	2	C–	18	255°	5	C+
7	90°	2	C–	19	270°	5	C+
8	105°	2	C–	20	285°	5	C+
9	120°	3	B+	21	300°	6	B–
10	135°	3	B+	22	315°	6	B–
11	150°	3	B+	23	330°	6	B–
12	165°	3	B+	24	345°	6	B–
Pole 1 (0°–165°)				Pole 2 (180°–345°)			

1.4 Step 3 – Lower Layer Assignment

1.4.1 Full-Pitched Winding ($y = 12$)

Three representative slots are shown; all 24 follow the same logic:

Slot j	Layer-1 of slot $j - 12 \Rightarrow$ Layer-2 of slot j
1	Layer-1[13] = A– A+
5	Layer-1[17] = C+ C–
13	Layer-1[1] = A+ A–

Slot:	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Layer 1:		A+				C–				B+				A–				C+				B–		
Layer 2:		A+				C–				B+				A–				C+				B–		

The coil pitch arrow: $\underbrace{1 \xrightarrow{y=12} 13}_{\text{Phase A coil 1}}, \underbrace{2 \rightarrow 14}_{\text{coil 2}}, \dots$

1.4.2 11/12 Short-Pitched Winding ($y = 11$)

Now $y = 11$: the angle difference between slot j and slot $j - 11$ is $11 \times 15^\circ = 165^\circ$ (not 180°). At group boundaries the phasor falls in the adjacent sector instead of the opposite one, creating *mixed-phase* slots.

Slot:	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Layer 1:		A+				C–				B+				A–				C+				B–		
Layer 2:	A+	A+	A+	C–	C–	C–	B+	B+	B+	A–	A–	A–	C+	C+	C+	B–	B–	B–	A+	A+	A+	A+	A+	A+

The highlighted (yellow) slots (4, 8, 12, 16, 20, 24) are the *boundary slots* where two different phases share the same slot. They arise because shortening the pitch by 1 slot shifts the return conductor out of the purely opposite sector by $180^\circ - 165^\circ = 15^\circ$ (one slot), crossing into the adjacent group.

1.5 Step 4 – Distribution Factor k_d

1.5.1 Physical Meaning

In a concentrated winding (all q coils of one phase under one pole placed in a single slot) the coil EMFs are perfectly in phase and add arithmetically. In a *distributed* winding the q coils are spread over q adjacent slots, each displaced by α_e electrically. Their EMF phasors form an arc of a regular polygon, and the vector sum is smaller than the arithmetic sum.

$$k_d = \frac{\text{vector (phasor) sum of the } q \text{ coil EMFs}}{\text{arithmetic sum of the } q \text{ coil EMFs}} \quad (5)$$

1.5.2 Derivation

Each of the q phasors has magnitude E_m and successive phasors are offset by α_e . The vector sum of q unit phasors starting at 0° and stepping by α_e :

$$\left| \sum_{i=0}^{q-1} e^{j i \alpha_e} \right| = \frac{\sin(q \alpha_e / 2)}{\sin(\alpha_e / 2)} \quad (6)$$

(standard result for a finite geometric series on the unit circle). The arithmetic sum is simply q . Hence:

$$k_{d,n} = \frac{\sin\left(\frac{n q \alpha_e}{2}\right)}{q \sin\left(\frac{n \alpha_e}{2}\right)} \quad (7)$$

where n is the harmonic order. For the n -th harmonic the electrical angle between adjacent phasors becomes $n \alpha_e$, so α_e is replaced by $n \alpha_e$ throughout.

1.5.3 Calculations for $q = 4$, $\alpha_e = 15^\circ$

Fundamental ($n = 1$):

$$k_{d,1} = \frac{\sin\left(\frac{1 \times 4 \times 15^\circ}{2}\right)}{4 \sin\left(\frac{1 \times 15^\circ}{2}\right)} = \frac{\sin 30^\circ}{4 \sin 7.5^\circ} = \frac{0.5000}{4 \times 0.1305} = \frac{0.5000}{0.5221} = \mathbf{0.9577} \quad (8)$$

Third harmonic ($n = 3$):

$$k_{d,3} = \frac{\sin\left(\frac{3 \times 4 \times 15^\circ}{2}\right)}{4 \sin\left(\frac{3 \times 15^\circ}{2}\right)} = \frac{\sin 90^\circ}{4 \sin 22.5^\circ} = \frac{1.0000}{4 \times 0.3827} = \frac{1.0000}{1.5307} = \mathbf{0.6533} \quad (9)$$

Fifth harmonic ($n = 5$):

$$k_{d,5} = \frac{\sin\left(\frac{5 \times 4 \times 15^\circ}{2}\right)}{4 \sin\left(\frac{5 \times 15^\circ}{2}\right)} = \frac{\sin 150^\circ}{4 \sin 37.5^\circ} = \frac{0.5000}{4 \times 0.6088} = \frac{0.5000}{2.4351} = \mathbf{0.2053} \quad (10)$$

The distribution factor *always decreases with harmonic order* for a fixed q – distributing the winding inherently attenuates higher harmonics.

1.6 Step 5 – Pitch Factor k_p

1.6.1 Physical Meaning

Consider one coil with its go side in slot k and its return side in slot $k + y$. The EMF induced in the go-side conductor is $e_{go}(t) = E_m \sin(\omega t)$. The EMF induced in the return-side conductor is $e_{ret}(t) = E_m \sin(\omega t - \theta_c)$ where $\theta_c = y \alpha_e$ is the electrical angle spanned by the coil. Because the return conductor is wound backwards (it is the return path of the coil), its contribution to the *net* coil EMF is $-e_{ret}$. The total coil EMF is:

$$e_{coil} = e_{go} - e_{ret} = E_m [\sin \omega t - \sin(\omega t - \theta_c)] = 2E_m \cos\left(\omega t - \frac{\theta_c}{2}\right) \sin \frac{\theta_c}{2} \quad (11)$$

The amplitude is $2E_m \sin(\theta_c/2)$. For a *full-pitched* coil $\theta_c = 180^\circ$, so the amplitude is $2E_m \sin(90^\circ) = 2E_m$ (maximum possible). The *pitch factor* is the ratio of the short-pitch amplitude to the full-pitch amplitude:

$$k_p = \frac{2E_m \sin(\theta_c/2)}{2E_m \sin(90^\circ)} = \sin \frac{\theta_c}{2} = \sin \frac{y \alpha_e}{2} = \sin \frac{y \pi}{2\tau_p} \quad (12)$$

For the n -th harmonic the field rotates n times faster, so the coil spans n times more electrical angle:

$$k_{p,n} = \sin \left(\frac{n y \pi}{2 \tau_p} \right) \quad (13)$$

1.6.2 Calculations

Full-pitched ($y = 12, \tau_p = 12$):

$$k_{p,1} = \sin \frac{1 \times 12 \pi}{2 \times 12} = \sin \frac{\pi}{2} = \mathbf{1.0000} \quad (14)$$

$$k_{p,3} = \left| \sin \frac{3 \times 12 \pi}{2 \times 12} \right| = \left| \sin \frac{3\pi}{2} \right| = |-1| = \mathbf{1.0000} \quad (15)$$

$$k_{p,5} = \left| \sin \frac{5 \times 12 \pi}{2 \times 12} \right| = \left| \sin \frac{5\pi}{2} \right| = |+1| = \mathbf{1.0000} \quad (16)$$

All harmonics have $k_p = 1$ – a full-pitched coil gives maximum EMF for every harmonic.

11/12 Short-pitched ($y = 11, \tau_p = 12$):

$$k_{p,1} = \sin \frac{1 \times 11 \pi}{2 \times 12} = \sin \frac{11\pi}{24} = \sin 82.5^\circ = \mathbf{0.9914} \quad (17)$$

$$k_{p,3} = \left| \sin \frac{3 \times 11 \pi}{2 \times 12} \right| = \left| \sin \frac{33\pi}{24} \right| = |\sin 247.5^\circ| = |-\sin 67.5^\circ| = \mathbf{0.9239} \quad (18)$$

$$k_{p,5} = \left| \sin \frac{5 \times 11 \pi}{2 \times 12} \right| = \left| \sin \frac{55\pi}{24} \right| = |\sin 412.5^\circ| = \sin 52.5^\circ = \mathbf{0.7934} \quad (19)$$

1.7 Step 6 – Winding Factor k_w

The winding factor combines both effects:

$$k_{w,n} = k_{d,n} \times k_{p,n} \quad (20)$$

Table 3: Complete winding factor results – Full-pitched vs. 11/12 short-pitched

n	Full-pitched ($y = 12$)			11/12 Short-pitched ($y = 11$)			Change	
	$k_{d,n}$	$k_{p,n}$	$k_{w,n}$	$k_{d,n}$	$k_{p,n}$	$k_{w,n}$	Δk_w	$\Delta\%$
1 (fund.)	0.9577	1.0000	0.9577	0.9577	0.9914	0.9495	-0.0082	-0.9%
3	0.6533	1.0000	0.6533	0.6533	0.9239	0.6036	-0.0497	-7.6%
5	0.2053	1.0000	0.2053	0.2053	0.7934	0.1629	-0.0424	-20.6%

1.8 Step 7 – Winding Diagram (One Pole-Pair)

Table 4 shows the complete double-layer slot layout for one pole-pair (slots 1–24). The symbol $\bullet$ means current flows *into* the page; $\times$ means current flows *out of* the page. The coil pitch arrow $k \rightarrow k + y$ connects each go-side (Layer 1) to its return-side (Layer 2).

Table 4: Winding diagram – full-pitched ($y = 12$), one pole-pair

Slot k	Pole 1												Pole 2												
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	
Layer 1			A●				C×				B●				A×				C●				B×		
Layer 2			A●				C×				B●				A×				C●				B×		
Coil span	$y = 12$ slots																								

Table 5: Winding diagram – 11/12 short-pitched ($y = 11$), one pole-pair. Yellow cells = boundary slots with mixed-phase layers.

Slot k	Pole 1												Pole 2												
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	
Layer 1			A●				C×				B●			A×					C●				B×		
Layer 2	A●	A●	A●	C×	C×	C×	C×	B●	B●	B●	B●	A×	A×	A×	A×	C●	C●	C●	C●	B×	B×	B×	B×	A●	A●
Coil span	$y = 11$ slots																								

1.9 Discussion and Comments

1. **Distribution factor decreases with harmonic order.** For $q = 4$ and $\alpha_e = 15^\circ$, going from $n = 1$ to $n = 5$ drops k_d from 0.958 to only 0.205. Spreading the winding over q slots is therefore a powerful tool for harmonic attenuation, even before any pitch adjustment.
2. **Full-pitched winding: pitch factor is unity for all odd harmonics.** Because $y\alpha_e = 180^\circ$, the formula $k_{p,n} = \sin(n\pi/2)$ gives ± 1 for every odd n . This means the full-pitched winding provides *no* pitch-based harmonic suppression – the 3rd harmonic winding factor of 0.653 is entirely due to k_d and remains high.
3. **Short-pitching disproportionately suppresses higher harmonics.** Shortening by just 1 slot ($y = 11$) changes $k_{p,1}$ by only -0.9% but reduces $k_{p,5}$ by -20.6% . The mathematical reason: $k_{p,n} = \sin(n \cdot y\pi/2\tau_p)$ is more sensitive to changes in y for larger n because the argument is multiplied by n . A small shift in y sweeps a larger angle for the 5th harmonic than for the fundamental.
4. **The 3rd harmonic is a triplen harmonic.** In a balanced 3-phase machine, triplen ($n = 3, 9, 15, \dots$) space harmonics do not produce average torque because they are in phase across all three phases (no 120° offset between phases for $n = 3$). However, they still cause iron losses and noise.
5. **Summary recommendation:** The 11/12 short-pitched winding is the better design. It costs $< 1\%$ of useful torque capability while achieving significant attenuation of harmful harmonics. In industrial practice, short pitching is almost universally applied in distributed-winding AC machines.

2 Q2 – Fractional-Slot Winding Design

2.1 Machine Parameters and Notation

Table 6: Design 1 parameters ($2575421 \bmod 5 = 1$)

Parameter	Symbol	Value
Number of stator slots	N_s	15
Number of poles (magnets)	N_m	8
Number of pole pairs	p	4
Phases	m	3
R_{ro}/R_{so} ratio	—	0.6

The slots per pole per phase is:

$$q = \frac{N_s}{N_m \cdot m} = \frac{15}{8 \times 3} = 0.625 \notin \mathbb{Z}^+ \quad (21)$$

Because q is not an integer, the simple coil-group rule of integral-slot windings (“ q consecutive slots per phase per pole”) does not apply, and the closed-form formulas for k_d and k_p are invalid. Instead, the *star of slots* phasor method must be used.

2.2 Step 1 – Electrical Angle per Slot α_s

For a machine with N_m poles the field repeats every pole pair (360° electrical). The electrical angle between two adjacent slots is:

$$\alpha_s = \frac{N_m \times 180^\circ}{N_s} = \frac{8 \times 180^\circ}{15} = \mathbf{96^\circ} \quad (22)$$

Note that $\alpha_s = 96^\circ > 60^\circ$: each step jumps *across* a sector boundary rather than staying within one 60° phase group. This is the signature of a fractional-slot winding – the slot phasors do not cluster neatly into phase groups.

2.3 Step 2 – Slot Phasor Angles (Star of Slots)

For a fractional-slot winding the winding factor cannot be obtained from simple closed-form expressions; the *star of slots* (phasor diagram) method is used instead. Each slot k is assigned an EMF phasor whose electrical angle is:

$$\varphi_k = (k - 1) \alpha_s \pmod{360^\circ} \quad (23)$$

The N_s phasors so generated form the star of slots. The fundamental winding factor of each phase is, by definition, the magnitude of the phasor sum of its coil-sides divided by the number of coil-sides, so to maximise $k_{w,1}$ each slot should be given to the phase whose axis its phasor most closely aligns with (with the sign chosen so that the contribution

adds rather than cancels). In practice this is implemented by the 60° -sector rule: the 360° circle is split into six 60° sectors labelled $+A$, $-C$, $+B$, $-A$, $+C$, $-B$, and each slot is assigned to the phase/sign of the sector its phasor falls into – which is exactly the assignment that makes every coil-side contribute its maximum possible projection onto that phase's axis. Table 7 lists the calculation for all 15 slots. Because $\gcd(15, 4) = 1$, no two slots share the same angle – all 15 phasors are distinct and the sequence visits 15 different positions before returning to 0° .

Table 7: Slot phasor angles – Design 1 ($\alpha_s = 96^\circ$)

Slot k	$(k-1) \times 96^\circ$	φ_k	Sector	Slot k	$(k-1) \times 96^\circ$	φ_k	Sector
1	0	0°	1	9	768	48°	1
2	96	96°	2	10	864	144°	3
3	192	192°	4	11	960	240°	5
4	288	288°	5	12	1056	336°	6
5	384	24°	1	13	1152	72°	2
6	480	120°	3	14	1248	168°	3
7	576	216°	4	15	1344	264°	5
8	672	312°	6				

2.4 Step 3 – Phase Assignment (60° Sector Rule)

The same six-sector rule from Q1 (Section 1) applies here. The 360° circle is divided into six 60° sectors, and each slot is assigned to the phase whose sector contains its phasor:

Sector	Angle range	Assignment
1	$[0^\circ, 60^\circ)$	A+
2	$[60^\circ, 120^\circ)$	C–
3	$[120^\circ, 180^\circ)$	B+
4	$[180^\circ, 240^\circ)$	A–
5	$[240^\circ, 300^\circ)$	C+
6	$[300^\circ, 360^\circ)$	B–

Applying this to Table 7 gives the full per-slot assignment in Table 8.

Table 8: Complete phase assignment – all 15 slots

Slot	φ_k	Sector	Assignment	Slot	φ_k	Sector	Assignment
1	0°	1	A+	9	48°	1	A+
2	96°	2	C–	10	144°	3	B+
3	192°	4	A–	11	240°	5	C+
4	288°	5	C+	12	336°	6	B–
5	24°	1	A+	13	72°	2	C–
6	120°	3	B+	14	168°	3	B+
7	216°	4	A–	15	264°	5	C+
8	312°	6	B–				

The result is balanced: each phase receives exactly $15/3 = 5$ slots.

- **Phase A:** A+ at slots 1, 5, 9 and A– at slots 3, 7
- **Phase B:** B+ at slots 6, 10, 14 and B– at slots 8, 12
- **Phase C:** C+ at slots 4, 11, 15 and C– at slots 2, 13

Figure 1 shows all 15 slot phasors arranged in the star of slots, colour-coded by phase.

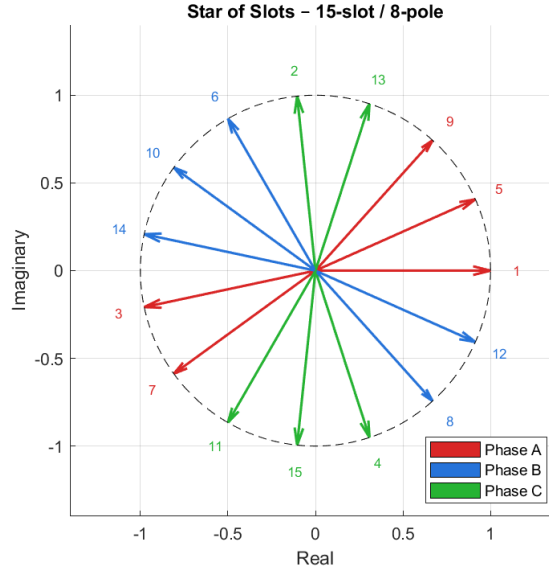


Figure 1: Star of slots for Design 1 ($N_s = 15$, $N_m = 8$). Colours: red = Phase A, blue = Phase B, green = Phase C. Solid arrows = positive direction; dashed = negative direction.

2.5 Step 4 – Winding Factor by Phasor Summation

2.5.1 Calculation Method

For the n -th space harmonic, each slot assigned to Phase A contributes one unit phasor. The general expression for the n -th harmonic winding factor by phasor summation is:

$$k_{w,n} = \frac{1}{N_{ph}} \left| \sum_{k \in \text{ph}^+} e^{jn\varphi_k} + \sum_{k \in \text{ph}^-} e^{j(n\varphi_k + \pi)} \right| \quad (24)$$

where N_{ph} is the number of slots per phase, and ph^+ , ph^- are the positive and negative slot sets of that phase. The angle of each contributing phasor therefore depends on whether the slot is positive or negative:

$$\text{A+ slot } k: e^{jn\varphi_k}, \quad \text{A- slot } k: e^{j(n\varphi_k + 180^\circ)} \quad (25)$$

The winding factor is then the magnitude of the vector sum divided by the number of slots per phase ($N_{ph} = 5$):

$$k_{w,n} = \frac{|\Sigma_A|}{N_{ph}} = \frac{|\Sigma_A|}{5} \quad (26)$$

The A– slots effectively have their phasors *reversed* (rotated 180°) before summing, which accounts for the fact that the current flows in the opposite direction in those slots.

2.5.2 Fundamental ($n = 1$)

Table 9: Phase A phasor contributions, $n = 1$

Slot	Type	φ_k	Angle used ($n\varphi_k + \text{rev.}$)	Real	Imag.
1	A+	0°	0°	+1.000	+0.000
5	A+	24°	24°	+0.914	+0.407
9	A+	48°	48°	+0.669	+0.743
3	A–	192°	$192^\circ + 180^\circ = 12^\circ$	+0.978	+0.208
7	A–	216°	$216^\circ + 180^\circ = 36^\circ$	+0.809	+0.588
Sum Σ_A				+4.370	+1.946

$$|\Sigma_A| = \sqrt{4.370^2 + 1.946^2} = \sqrt{19.10 + 3.79} = 4.783 \quad (27)$$

$$k_{w,1} = \frac{4.783}{5} = 0.9566 \quad (28)$$

Observation: After reversing the A– phasors (slots 3 and 7), all five phasors cluster in the first quadrant within a 48° arc. They nearly align, which explains the high winding factor.

2.5.3 Third Harmonic ($n = 3$)

For $n = 3$ the phasor angle of each slot is multiplied by 3 before applying the reversal for A– slots.

Table 10: Phase A phasor contributions, $n = 3$

Slot	Type	φ_k	Angle used ($3\varphi_k + \text{rev.}$)	Real	Imag.
1	A+	0°	0°	+1.000	+0.000
5	A+	24°	72°	+0.309	+0.951
9	A+	48°	144°	–0.809	+0.588
3	A–	192°	$576^\circ + 180^\circ = 36^\circ$	+0.809	+0.588
7	A–	216°	$648^\circ + 180^\circ = 108^\circ$	–0.309	+0.951
Sum Σ_A				+1.000	+3.078

$$|\Sigma_A| = \sqrt{1.000^2 + 3.078^2} = \sqrt{1.000 + 9.474} = 3.236 \quad (29)$$

$$k_{w,3} = \frac{3.236}{5} = 0.6472 \quad (30)$$

2.5.4 Fifth Harmonic ($n = 5$)

Table 11: Phase A phasor contributions, $n = 5$

Slot	Type	φ_k	Angle used ($5\varphi_k + \text{rev.}$)	Real	Imag.
1	A+	0°	0°	+1.000	+0.000
5	A+	24°	120°	−0.500	+0.866
9	A+	48°	240°	−0.500	−0.866
3	A−	192°	$960^\circ + 180^\circ = 60^\circ$	+0.500	+0.866
7	A−	216°	$1080^\circ + 180^\circ = 180^\circ$	−1.000	+0.000
Sum Σ_A				−0.500	+0.866

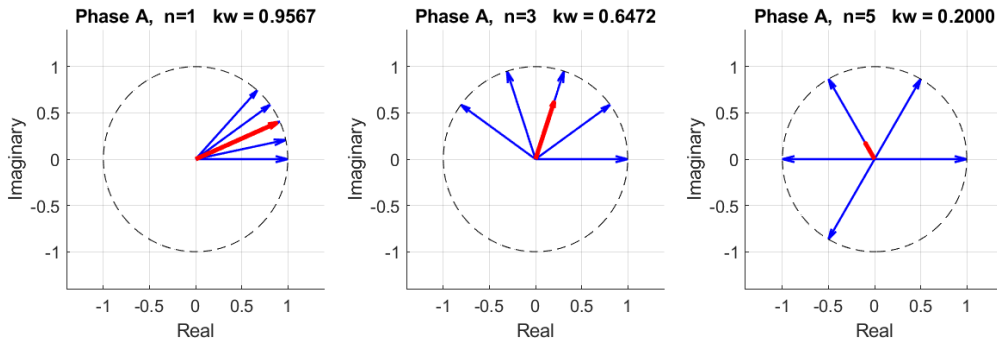
$$|\Sigma_A| = \sqrt{(-0.500)^2 + (0.866)^2} = \sqrt{0.250 + 0.750} = 1.000 \quad (31)$$

$$k_{w,5} = \frac{1.000}{5} = 0.200 \quad (32)$$

Table 12: Summary of winding factors – Design 1 (15-slot/8-pole)

Harmonic	Σ_A	$ \Sigma_A $	$k_{w,n}$	Note
$n = 1$ (fund.)	$4.370 + j1.946$	4.783	0.9566	High, good winding configuration
$n = 3$ (triplen)	$1.000 + j3.078$	3.236	0.6472	Zero-seq. no net torque
$n = 5$	$-0.500 + j0.866$	1.000	0.2000	Suppressed

Figure 2 shows the Phase A phasor diagrams for all three harmonics. Each arrow is one slot phasor (after reversing A− slots); the thick red arrow is the vector resultant.

Figure 2: Phase A phasor diagrams for $n = 1, 3, 5$.

Convention note. Each tooth-coil is represented by a single Layer-1 phasor at its tooth-centre angle (return-sides not summed separately), so per Hanselman $k_{w,n} = |\sum_{i=1}^5 e^{jn\varphi_i}|/5$.

2.6 Step 5 – Constructing the Winding Diagram

The star of slots provides one phase assignment per slot, which is taken as the upper layer (Layer 1) of each slot. In this concentrated tooth-coil topology each coil is wound around a single tooth, so its go-side sits in Layer 1 of slot k and its return-side in Layer 2 of slot $k + 1$. Because the return-side carries the opposite current direction, the lower layer of any slot is simply the negative of the upper layer of the preceding slot.

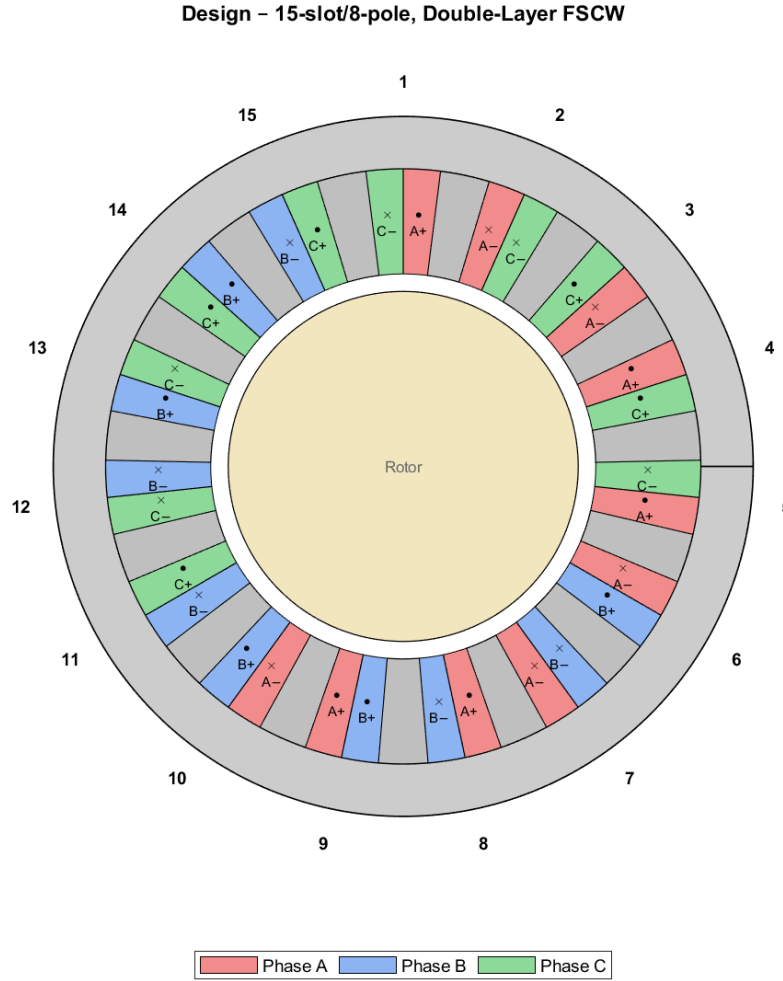


Figure 3: Winding layout for Design 1 (15-slot/8-pole, 3-phase).

2.7 Discussion

1. **High fundamental winding factor for a concentrated winding.** $k_{w,1} = 0.9566$ is comparable to that of distributed windings. The 15/8 combination is identified in the Hanselman book as one of the best fractional-slot combinations for torque density, and the value $k_{w,1} = 0.9566$ obtained here matches the example tabulated in the book.

2. **Natural suppression of the 5th harmonic.** The three A+ slots contribute phasors at 0° , 120° , 240° for $n = 5$ – a balanced triphasic set that sums to zero. Only the A– slots restore a small residual ($|\Sigma| = 1.0$), giving $k_{w,5} = 0.2$. This suppression is geometric and requires no pitch adjustment.
3. **3rd harmonic: large winding factor, zero net rotating field.** $k_{w,3} = 0.647$ is large, but as discussed in Q1, the 3rd space harmonic is triplen: balanced condition $I_A + I_B + I_C = 0$ cancels the net rotating component. It may still cause iron losses through standing-wave effects.
4. **Two phases share each slot.** Unlike integral-slot windings where each slot is occupied by a single phase, the double-layer FSCW places two *different* phases in each slot. Thicker Insulation is needed.
5. **Short end-windings – key FSCW advantage.** Because each coil spans only one tooth pitch (approximately $360^\circ/(N_s) = 24^\circ$ mechanical), the overhang is very short. This reduces copper volume, DC resistance, and copper losses.

3 Q3 – Analytical Modelling

3.1 Machine Dimensions

From the assignment common specifications and the Design 1 ratio $R_{ro}/R_{so} = 0.6$. This split between rotor and stator is not a free design parameter; the Common Specs require that “the rotor outside radius R_m is chosen to maximise the motor constant $K_m = T/\sqrt{P_{Cu}}$.” For surface-PM machines the motor constant can be written $K_m \propto B_g \alpha_m R_{ro} \sqrt{A_{slot}}$, where the slot area itself depends on the radial budget left over for the slots, $R_{so} - (R_{ro} + g + w_{sy})$. Differentiating with respect to R_{ro} and equating to zero gives an optimum split that, for the assumed loadings and a magnet thickness of $l_m = 4$ mm, falls near $R_{ro}/R_{so} \approx 0.55$ – 0.65 ; Hanselman tabulates the precise value of 0.60 for Design 1 of the 15-slot/8-pole family. The geometry inputs that follow take this as given:

Table 13: Machine geometry – Design 1

Parameter	Symbol	Value
Stator outer radius	R_{so}	50.0 mm
Rotor outer radius	$R_{ro} = 0.6 R_{so}$	30.0 mm
Air-gap length	g	1.0 mm
Magnet radial thickness	l_m	4.0 mm
Stator inner radius	$R_{si} = R_{ro} + g$	31.0 mm
Rotor core outer radius	$R_{ro,c} = R_{ro} - l_m$	26.0 mm
Stack length	L_s	100 mm
Magnet remanence	B_r	1.3 T
Magnet relative permeability	μ_r	1.05
Magnet arc ratio	α_m	0.889
Target iron flux density	B_t	1.4 T
Current density (rms)	J_{rms}	5 A/mm ²
Slot fill factor	k_{wc}	0.60

3.2 Slot Pitch and Tooth Geometry

The slot pitch τ_s is the circumferential distance between the centrelines of two adjacent slots, measured at the stator bore:

$$\tau_s = \frac{2\pi R_{si}}{N_s} = \frac{2\pi \times 31}{15} = 12.99 \text{ mm} \quad (33)$$

where R_{si} is the stator inner radius and N_s the total slot count. Figure 4 shows the relevant geometry in a linearised (developed) view of one slot pitch.

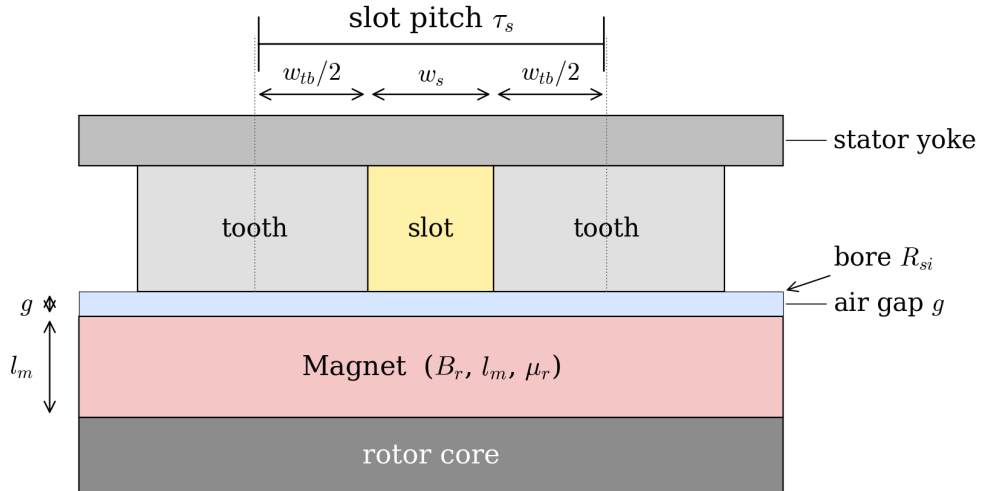


Figure 4: Developed (unrolled) view of the stator bore showing one slot pitch τ_s , the tooth body width w_{tb} (split half-and-half between the two neighbouring teeth), the slot opening w_s , the air gap g , and the surface magnet of thickness l_m .

A first-pass estimate of the peak air-gap flux density, ignoring the slotting effect ($k_c =$

1), follows from a one-dimensional magnetic equivalent circuit of a surface-PM rotor with a smooth stator. Assuming the stator iron is infinitely permeable (no H -drop in the steel) and that the magnet and air-gap cross-sections are equal so that $B_m = B_g$ by flux continuity, two relations govern the operating point:

- Magnet B–H line (linearised recoil curve): $B_m = B_r + \mu_0 \mu_r H_m \implies H_m = (B_m - B_r)/(\mu_0 \mu_r)$.
- Ampère’s law around one pole loop (magnet $\rightarrow$ gap $\rightarrow$ iron $\rightarrow$ gap $\rightarrow$ magnet, iron contributes nothing): $H_m l_m + H_g g = 0$, with $H_g = B_g/\mu_0$.

Substituting the B–H line into Ampère’s law and using $B_m = B_g$:

$$\frac{B_g - B_r}{\mu_0 \mu_r} l_m = -\frac{B_g}{\mu_0} g \quad (34)$$

Multiplying through by μ_0 , expanding, and collecting the B_g terms gives $B_g (l_m/\mu_r + g) = B_r l_m/\mu_r$, which rearranges to:

$$B_{g,0} = \frac{B_r l_m}{l_m + \mu_r g} = \frac{1.3 \times 4}{4 + 1.05 \times 1} = 1.030 \text{ T} \quad (35)$$

Equivalently, the magnet behaves as a flux source B_r in series with its own reluctance $\mathcal{R}_m = l_m/(\mu_0 \mu_r A)$, loaded by the gap reluctance $\mathcal{R}_g = g/(\mu_0 A)$, and B_g/B_r is the reluctance-divider ratio $\mathcal{R}_m/(\mathcal{R}_m + \mathcal{R}_g) = l_m/(l_m + \mu_r g)$. Figure 5 shows the corresponding magnetic equivalent circuit. With Carter’s coefficient included (next subsection), g is replaced by the effective gap $k_c g$, giving the corrected $B_{g,\text{peak}} = B_r l_m/(l_m + \mu_r k_c g)$ used below.

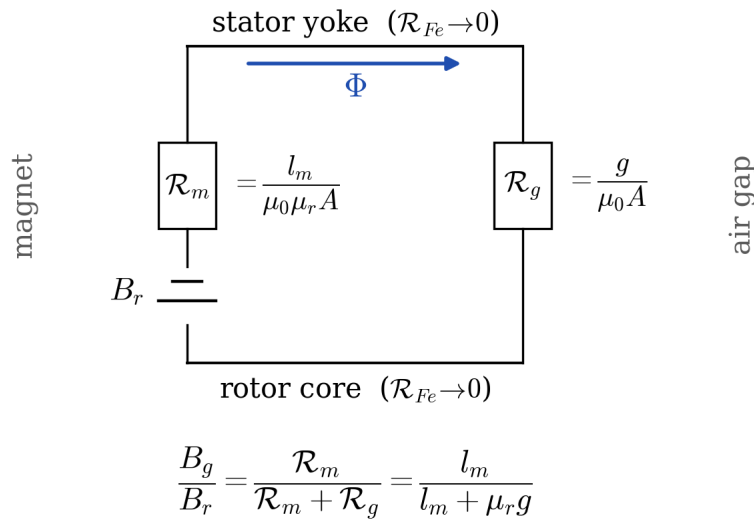


Figure 5: One-dimensional magnetic equivalent circuit of the surface-PM rotor. The magnet is represented by the MMF source B_r in series with its reluctance $\mathcal{R}_m$; the air gap by $\mathcal{R}_g$; the iron pieces by ideal shorts. B_g/B_r is the reluctance-divider ratio $\mathcal{R}_m/(\mathcal{R}_m + \mathcal{R}_g) = l_m/(l_m + \mu_r g)$.

The corresponding average flux density seen by one tooth – the magnet arc spans only a fraction α_m of the pole pitch – is:

$$B_{g,\text{avg}} = B_{g,0} \alpha_m = 1.030 \times 0.889 = 0.916 \text{ T} \quad (36)$$

Conservation of flux requires the flux entering the air gap over one slot pitch to be funnelled through one tooth of width w_{tb} (Figure 6). Setting the tooth flux density equal to the target value B_t gives:

$$B_{g,\text{avg}} \cdot \tau_s = B_t \cdot w_{tb} \implies w_{tb} = \frac{B_{g,\text{avg}} \tau_s}{B_t} = \frac{0.916 \times 12.99}{1.4} = 8.50 \text{ mm} \quad (37)$$

The remaining width of the slot pitch is taken as the slot opening:

$$w_s = \tau_s - w_{tb} = 12.99 - 8.50 = 4.49 \text{ mm} \quad (38)$$

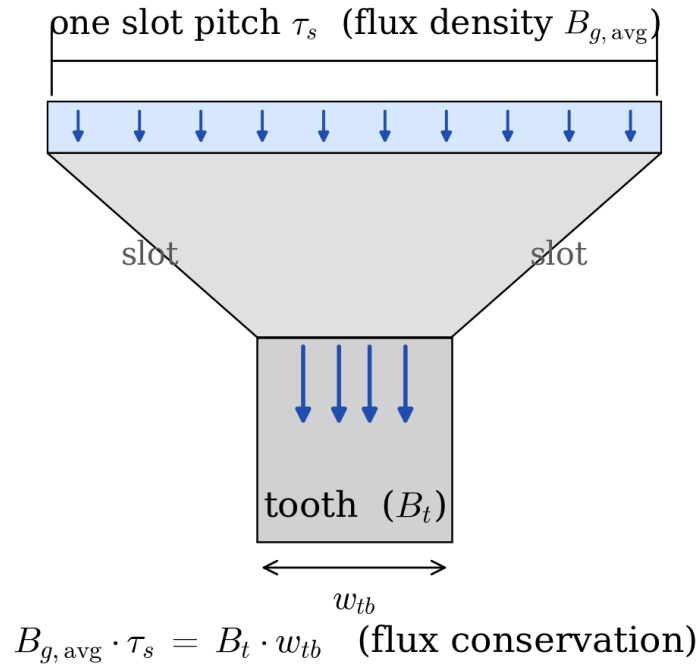


Figure 6: Flux funnelling. The air-gap flux entering the stator over one slot pitch τ_s at average density $B_{g,\text{avg}}$

3.3 Carter's Coefficient

Stator slotting causes the air-gap flux density to dip above each slot opening, which raises the average reluctance of the gap. Carter's coefficient $k_c \geq 1$ captures this effect by replacing the slotted stator with an equivalent smooth surface having an enlarged air gap $k_c g$.

$$k_c = \frac{\tau_s}{\tau_s - \kappa w_s}, \quad \kappa = \frac{w_s/g}{5 + w_s/g} \quad (39)$$

The symbols are:

- $\tau_s = 2\pi R_{si}/N_s$ – slot pitch
- w_s – slot opening width

- g – air-gap length;
- κ – dimensionless slot-opening factor

Applying Eq. (39) with $g = 1$ mm and $w_s = 4.49$ mm:

$$\frac{w_s}{g} = 4.49, \quad \kappa = \frac{4.49}{5 + 4.49} = \frac{4.49}{9.49} = 0.473 \quad (40)$$

$$k_c = \frac{12.99}{12.99 - 0.473 \times 4.49} = \frac{12.99}{12.99 - 2.124} = \frac{12.99}{10.866} = \mathbf{1.195} \quad (41)$$

3.4 Effective Air Gap and Corrected Flux Density

For a surface-mount PM machine with radially magnetised magnets the rotor surface and stator bore are treated as concentric cylinders, and the peak air-gap flux density is:

$$B_{g,\text{peak}} = \frac{B_r l_m}{l_m + \mu_r k_c g} \quad (42)$$

where B_r is the magnet remanence, l_m the magnet radial thickness, μ_r the magnet relative permeability, g the physical air gap, and k_c Carter's coefficient. The effective air gap including the magnet contribution is:

$$g_{\text{eff}} = k_c g + \frac{l_m}{\mu_r} = 1.195 \times 1 + \frac{4}{1.05} = 1.195 + 3.810 = 5.005 \text{ mm} \quad (43)$$

Substituting numerical values:

$$B_{g,\text{peak}} = \frac{1.3 \times 4}{4 + 1.05 \times 1.195 \times 1} = \frac{5.2}{5.255} = \mathbf{0.989 \text{ T}} \quad (44)$$

3.5 Effective Axial Length

Fringing at the axial ends of the machine extends the flux-linking length beyond the physical stack length L_s . A common approximation is:

$$L_e = L_s + 2g \quad (45)$$

Numerically:

$$L_e = 100 + 2 \times 1 = \mathbf{102 \text{ mm}} \quad (46)$$

3.6 Air-Gap Peak Flux Density Summary

All three values are consistent: the slotting correction reduces $B_{g,\text{peak}}$ from 1.030 T (idealised) to 0.989 T (with Carter correction), a reduction of 3.9%.

3.7 Flux Per Pole

The flux per pole, needed to size the back-iron and to calculate the induced EMF, is:

$$\Phi_{pole} = B_{g,peak} \alpha_m \tau_p L_e, \quad \tau_p = \frac{\pi R_{si}}{p} \quad (47)$$

where $\alpha_m = \theta_m/180^\circ$ is the magnet arc ratio ($160^\circ/180^\circ = 0.889$ here) and τ_p is the pole pitch at the stator bore. Evaluating:

$$\tau_p = \frac{\pi \times 31}{4} = 24.35 \text{ mm} \quad (48)$$

$$\Phi_{pole} = 0.989 \times 0.889 \times 24.35 \times 10^{-3} \times 102 \times 10^{-3} = \mathbf{2.18 \text{ mWb}} \quad (49)$$

3.8 Magnetic Loading

The magnetic loading $\bar{B}$ is the average value of the radial air-gap flux density taken over one full pole pitch. For a surface-mount PM machine with a magnet arc ratio α_m , the flux is essentially uniform under the magnet and (for the analytical model) zero between magnets, so:

$$\bar{B} = B_{g,peak} \cdot \alpha_m = 0.989 \times 0.889 = \mathbf{0.879 \text{ T}} \quad (50)$$

$\bar{B}$ together with the electrical loading A_{rms} determine the shear stress (and hence the torque) the machine can develop, as expressed by Eq. (62) below.

3.9 Iron Yoke Widths

The stator and rotor yoke widths are sized so that the peak flux density in the iron does not exceed the target $B_t = 1.4 \text{ T}$:

$$w_{sy} = w_{ry} = \frac{\Phi_{pole}}{2 B_t L_e} \quad (51)$$

Numerically:

$$w_{sy} = w_{ry} = \frac{2.18 \times 10^{-3}}{2 \times 1.4 \times 0.102} = \mathbf{7.64 \text{ mm}} \quad (52)$$

The rotor shaft radius is $R_{ro,c} - w_{ry} = 26 - 7.64 = 18.4 \text{ mm}$.

Self-consistency check on tooth and yoke flux density

The tooth and yoke widths above were sized using the pre-Carter estimate $B_{g,0} = 1.030 \text{ T}$. After the Carter correction the actual peak air-gap flux density drops to 0.989 T , so the iron flux densities should be re-checked. By flux conservation, the actual tooth flux density is

$$B_{tooth}^{actual} = \frac{B_{g,peak} \alpha_m \tau_s}{w_{tb}} = \frac{0.989 \times 0.889 \times 12.99}{8.50} \approx \mathbf{1.343 \text{ T}} \quad (53)$$

which sits just below the $B_t = 1.4\text{ T}$ target – the slotting correction provides a small natural margin in the teeth. The yoke flux density, found from the actual pole flux $\Phi_{pole} = 2.18\text{ mWb}$, is

$$B_{\text{yoke}}^{\text{actual}} = \frac{\Phi_{pole}}{2 w_{sy} L_e} = \frac{2.18 \times 10^{-3}}{2 \times 7.64 \times 10^{-3} \times 0.102} \approx \mathbf{1.398\text{ T}} \quad (54)$$

which lands essentially at the design target (this is by construction, since w_{sy} was sized for $B_t = 1.4\text{ T}$ with the Carter-corrected Φ_{pole}). The two cross-checks confirm the analytical design is self-consistent at the chosen 1.4 T limit.

3.10 Electrical Loading

Available slot height and slot area (rectangular approximation):

$$h_{slot} = (R_{so} - R_{si}) - w_{sy} = 19 - 7.64 = 11.36\text{ mm} \quad (55)$$

$$A_{slot} = w_s \times h_{slot} = 4.49 \times 11.36 = 51.0\text{ mm}^2 \quad (56)$$

The linear current density (electrical loading) characterises the ampere-conductor loading on the stator bore circumference:

$$A_{rms} = \frac{J_{rms} k_{wc} A_{slot}}{\tau_s} \quad (57)$$

where J_{rms} is the rms current density in the conductors, k_{wc} the slot fill factor, and A_{slot} the net slot cross-section. Substituting:

$$A_{rms} = \frac{5 \times 0.60 \times 51.0}{12.99} = \mathbf{11.78\text{ kA/m}} \quad (58)$$

Phase current, wire size and slot-fill verification

Anticipating $N_{cs} = 8$ conductors per slot from Section 3.13, the total ampere-conductors per slot is $N_{cs} I_{rms} = A_{rms} \tau_s$, so the phase RMS current is

$$I_{rms} = \frac{A_{rms} \tau_s}{N_{cs}} = \frac{11.78 \times 10^3 \times 12.99 \times 10^{-3}}{8} = \mathbf{19.1\text{ A}} \quad (59)$$

The required net copper area per conductor follows from the prescribed current density $J_{rms} = 5\text{ A/mm}^2$:

$$A_{\text{wire}} = \frac{I_{rms}}{J_{rms}} = \frac{19.1}{5} = \mathbf{3.83\text{ mm}^2} \quad (60)$$

The actual slot fill is then

$$k_{wc}^{\text{check}} = \frac{N_{cs} A_{\text{wire}}}{A_{slot}} = \frac{8 \times 3.83}{51.0} = 0.601 \quad (61)$$

which matches the assumed 0.60 by construction (since A_{rms} was derived from J_{rms} and k_{wc}); the check confirms that 8 conductors of the calculated wire area fit the slot without violating the fill-factor target. Since it is fractional slot there may be 2 different phases 4 wires for each.

3.11 Shear Stress and Expected Torque

Tangential force per unit air-gap surface area arises from the interaction between the radial flux density produced by the magnets and the tangential current sheet produced by the stator winding. Time-averaging over one electrical cycle, the electromagnetic shear stress at the stator bore is:

$$\sigma = \frac{1}{2} k_{w,1} B_{g,\text{peak}} A_{\text{peak}} = \frac{1}{\sqrt{2}} k_{w,1} B_{g,\text{peak}} A_{\text{rms}} \quad (62)$$

The fundamental winding factor $k_{w,1}$ accounts for the fact that only the fundamental space harmonic of the winding produces useful torque; the factor of $1/\sqrt{2}$ converts from peak to rms current loading. Integrating σ over the stator bore area $2\pi R_{si} L_e$ and multiplying by the lever arm R_{si} gives the average electromagnetic torque, and multiplying by the mechanical angular speed $\omega_m = 2\pi n/60$ gives the shaft power:

$$T = \sigma \cdot 2\pi R_{si}^2 L_e, \quad P = T \omega_m \quad (63)$$

Inserting the values computed above:

$$\sigma = \frac{1}{\sqrt{2}} \times 0.9566 \times 0.989 \times 11,780 = \mathbf{7.87 \text{ kPa}} \quad (64)$$

$$T = 7870 \times 2\pi \times (0.031)^2 \times 0.102 = \mathbf{4.85 \text{ N} \cdot \text{m}} \quad (65)$$

$$P = 4.85 \times \frac{1500 \times 2\pi}{60} = \mathbf{762 \text{ W}} \quad (66)$$

3.12 Specific Machine Constant

The classical output equation collects the magnetic loading $\hat{B} = B_{g,\text{peak}}$, the electric loading $\bar{A} = A_{\text{rms}}$, and the fundamental winding factor into a single figure-of-merit:

$$S_i = \frac{\pi^2}{\sqrt{2}} k_{w,1} \bar{A} \hat{B} D^2 L_e n_{\text{syn}} = C D^2 L_e n_{\text{syn}}, \quad C = \frac{\pi^2}{\sqrt{2}} k_{w,1} \bar{A} \hat{B} \quad (67)$$

The constant C depends only on the materials and the cooling capability, not on the volume; machines of different sizes can be compared at a glance through C alone.

For Design 1 with $\hat{B} = 0.989 \text{ T}$, $\bar{A} = 11.78 \text{ kA/m}$ and $k_{w,1} = 0.9566$:

$$C = \frac{\pi^2}{\sqrt{2}} \times 0.9566 \times 11,780 \times 0.989 \approx \mathbf{77.8 \text{ kW} \cdot \text{s/m}^3} \quad (68)$$

As a consistency check on the torque derived above, $D = 2R_{si} = 62 \text{ mm}$ and $n_{\text{syn}} = 1500/60 = 25 \text{ s}^{-1}$:

$$P_{\text{out}} \approx C D^2 L_e n_{\text{syn}} = 77,800 \times (0.062)^2 \times 0.102 \times 25 \approx 762 \text{ W}, \quad (69)$$

in agreement with the shear-stress-based result of the previous subsection. The value of $C \approx 78 \text{ kW} \cdot \text{s/m}^3$ is typical of a small, naturally-cooled industrial PMSM (Pyrhönen quotes 70–150 $\text{kW} \cdot \text{s/m}^3$ for this class of machines).

3.13 Number of Turns Per Phase

The number of series turns per phase is sized so that the back-EMF at rated speed equals the available phase voltage from the inverter. Assuming an SVPWM-driven three-phase voltage-source inverter on a DC bus $V_{dc} = 48\text{ V}$, the maximum peak phase voltage (line-to-neutral, fundamental component) in the linear modulation region is

$$\hat{V}_{ph} = \frac{V_{dc}}{\sqrt{3}} = \frac{48}{\sqrt{3}} \approx 27.7\text{ V} \quad (70)$$

The electrical angular frequency relates to the mechanical speed $n = 1500\text{ rpm}$ through the pole-pair count p :

$$\omega_e = p \omega_m = p \cdot \frac{2\pi n}{60} = 4 \times \frac{1500 \times 2\pi}{60} = 628.3\text{ rad/s} \quad (71)$$

For a sinusoidally distributed flux linkage, the peak back-EMF is $\hat{E}_{ph} = k_{w,1} N_{ph} \omega_e \Phi_{pole}$; equating this to $\hat{V}_{ph}$ and solving for N_{ph} :

$$N_{ph} = \frac{\hat{V}_{ph}}{k_{w,1} \omega_e \Phi_{pole}} = \frac{27.7}{0.9566 \times 628.3 \times 2.18 \times 10^{-3}} \approx 21.2 \quad (72)$$

where $k_{w,1}$ reduces the contribution of each turn to account for the spatial distribution of the conductors. The number of conductors per slot must be an integer, and is constrained by $N_{cs} = 2N_{ph}m/N_s$ for a double-layer winding. Rounding to the nearest valid integer:

$$N_{cs} = 8 \implies N_{ph} = \frac{N_{cs} N_s}{2m} = \frac{8 \times 15}{2 \times 3} = 20 \quad (73)$$

Back-EMF at 1500 rpm: $\hat{E}_{ph} = 0.9566 \times 628.3 \times 2.18 \times 10^{-3} \times 20 = 26.2\text{ V}$. This leaves a margin of $(27.7 - 26.2)/27.7 \approx 5.4\%$ to the inverter limit.

3.14 Phase Resistance and Copper Loss at 50°C

The Common Specs require the windings to be evaluated at $T_{Cu} = 50^\circ\text{C}$. Copper resistivity at this temperature is

$$\rho_{Cu}(50^\circ\text{C}) = \rho_{Cu}(20^\circ\text{C})(1 + \alpha \Delta T) = 1.72 \times 10^{-8} \times (1 + 0.00393 \times 30) \approx 1.92 \times 10^{-8} \Omega \cdot \text{m} \quad (74)$$

For the 15-slot/8-pole tooth-coil layout, each coil spans one tooth. The mean turn length is approximated as twice the axial length plus two end-turn arcs of one slot pitch each:

$$\ell_{\text{turn}} \approx 2L_e + 2\tau_s = 2 \times 102 + 2 \times 12.99 \approx 230\text{ mm} \quad (75)$$

With $N_{ph} = 20$ series turns per phase, the total series wire length is $L_{\text{wire}} = N_{ph} \ell_{\text{turn}} \approx 4.60\text{ m}$. The phase resistance is

$$R_{ph} = \frac{\rho_{Cu} L_{\text{wire}}}{A_{\text{wire}}} = \frac{1.92 \times 10^{-8} \times 4.60}{3.83 \times 10^{-6}} \approx \mathbf{23\text{ m}\Omega} \quad (76)$$

The total copper loss for the three phases at the rated RMS current is

$$P_{Cu} = 3 I_{rms}^2 R_{ph} = 3 \times (19.1)^2 \times 0.023 \approx \mathbf{25\text{ W}} \quad (77)$$

Treating this as the only loss (iron and stray losses neglected at this analytical stage), the rough efficiency at rated point is

$$\eta_{\text{approx}} \approx \frac{P_{\text{out}}}{P_{\text{out}} + P_{Cu}} = \frac{762}{762 + 25} \approx \mathbf{96.8\%}. \quad (78)$$

Adding a plausible 20–30 W of iron and stray losses would bring the efficiency to the 93–95% range.

3.15 Summary of Analytical Results

Table 14: Analytical modelling results – Design 1

Parameter	Symbol	Value
Slot pitch	τ_s	12.99 mm
Tooth width	w_{tb}	8.50 mm
Slot opening	w_s	4.49 mm
Carter coefficient	k_c	1.195
Effective air gap	g_{eff}	5.005 mm
Effective axial length	L_e	102 mm
Air-gap peak flux density	$B_{g,\text{peak}}$	0.989 T
Flux per pole	Φ_{pole}	2.18 mWb
Magnetic loading	$\bar{B}$	0.879 T
Stator yoke width	w_{sy}	7.64 mm
Rotor yoke width	w_{ry}	7.64 mm
Slot height	h_{slot}	11.36 mm
Electrical loading (rms)	A_{rms}	11.78 kA/m
Phase RMS current	I_{rms}	19.1 A
Conductor area	A_{wire}	3.83 mm ²
Tooth flux density (actual)	B_{tooth}	1.34 T
Yoke flux density (actual)	B_{yoke}	1.40 T
Shear stress	σ	7.87 kPa
Expected torque	T	4.85 N·m
Expected output power	P	762 W
Specific machine constant	C	77.8 kW·s/m ³
Turns per phase	N_{ph}	20
Conductors per slot	N_{cs}	8
Phase resistance (50°C)	R_{ph}	23 mΩ
Copper loss	P_{Cu}	25 W
Estimated efficiency	η	96.8%

4 Q4 – 2D FEA Modelling (ANSYS Maxwell)

4.1 Model Setup

4.1.1 Geometry

The 2D cross-section of Design 1 is shown in Figure 7. All dimensions are taken from the analytical design (Section 3) without optimisation. The stator is modelled as a full 360° disk because $\gcd(N_s, 2p) = \gcd(15, 8) = 1$, leaving no rotational symmetry that could be exploited with master/slave boundaries. Slots are drawn as simple rectangles (no pole shoes, per the project tip allowing simplified slot geometry). Magnets are modelled as annular sectors of 40° mechanical arc (160° electrical) at $R_{ro,c} \leq r \leq R_{ro}$, alternating radial magnetisation outward (N) and inward (S) around the rotor.

Design 1 -- 15-slot / 8-pole SPMSM cross-section

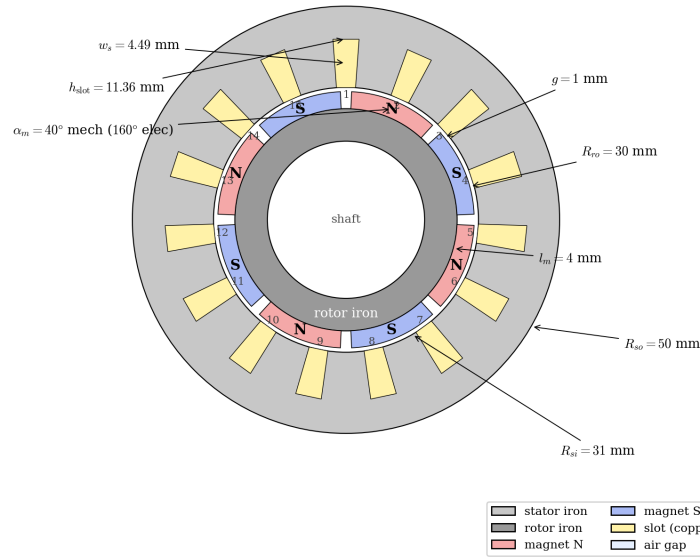


Figure 7: Cross-section of Design 1 modelled in ANSYS Maxwell 2D. 15 stator slots ($N_s = 15$, slot opening $w_s = 4.49$ mm, height $h_{\text{slot}} = 11.36$ mm), 8 surface-mount magnets ($N_m = 8$, arc 40° mech, $l_m = 4$ mm), air gap $g = 1$ mm, model depth $L_e = 102$ mm.

4.1.2 Materials

Table 15: Material assignments in the Maxwell 2D model.

Region	Library material	Notes
Stator iron	M19_24G_custom	non-linear B–H, see below
Rotor iron	M19_24G_custom	same as stator
Magnets ($\times 4$)	NdFeB_N (custom)	$H_c = -985\,000$ A/m
Magnets ($\times 4$)	NdFeB_S (custom)	$H_c = +985\,000$ A/m
Slot copper	Copper	current set by Coil Terminal
Air gap, outer air	Vacuum	—
Shaft	Stainless 304	magnetically inert

Iron laminations (M19_24G). Non-oriented Si-steel, 24-gauge (~ 0.355 mm lamination thickness). Cloned from the Maxwell library and replaced with the user-imported B–H curve in `HW2/M19_24G_BH.tab` (16-point dataset reaching $B = 2.05$ T at $H = 31,831$ A/m, knee near ~ 1.4 T). Lamination set v3 (stack axis along global Z , i.e. out of the drawing plane), stacking factor 0.95, bulk conductivity $\sigma = 2.083 \times 10^6$ S/m (only relevant for transient eddy-loss computations – in magnetostatic this value is unused), density 7650 kg/m³. The same assignment is used for both the stator annulus ($R_{si} = 31$ mm $\rightarrow$ $R_{so} = 50$ mm with the 15 slot cut-outs) and the rotor core ($r \leq R_{ro,c} = 26$ mm).

Permanent magnets (NdFeB). Sintered NdFeB with $B_r = 1.3$ T and $\mu_r = 1.05$, modelled as two project materials so that adjacent magnets can have opposite polarity while sharing a single per-magnet local CS direction:

$$H_c = \frac{B_r}{\mu_0 \mu_r} = \frac{1.3}{4\pi \times 10^{-7} \times 1.05} \approx 985,000 \text{ A/m}. \quad (79)$$

Each magnet has an Object Coordinate System (`CS_Mag01` . . . `CS_Mag08`) whose local $+X$ axis points radially outward through that magnet’s centre. The N material (`NdFeB_N`, $H_c = -985\,000$ A/m) places magnets 1, 3, 5, 7 with B_r along $+X$ (radially outward), and the S material (`NdFeB_S`, $H_c = +985\,000$ A/m) places magnets 2, 4, 6, 8 with B_r along $-X$ (radially inward), producing the required alternating N–S–N–S–N–S–N–S sequence around the rotor. Magnet bulk conductivity is set to zero (the eddy-current contribution of the magnets is irrelevant for the magnetostatic study).

4.1.3 Excitations, Boundary Conditions and Mesh

Excitations. Three windings (A, B, C) are defined. The 15 coils are assigned according to the star-of-slots result from Section 2: Phase A on teeth 1, 3, 5, 7, 9; Phase B on teeth 6, 8, 10, 12, 14; Phase C on teeth 2, 4, 11, 13, 15. Each Coil Terminal carries $N_{cs} = 8$ conductors.

Boundary conditions. A circular outer air region of radius $1.5 R_{so} = 75$ mm is added; vector potential $A_z = 0$ is enforced on its outer edge to contain the flux. Maxwell’s

“Model Depth” setting is set to $L_e = 102\text{ mm}$ so that 2D quantities (flux, torque, EMF) come out scaled by the effective axial length automatically.

Mesh. The air gap is split into two thin half-gap layers to apply mesh operation to increase the calculation accuracy in the air gap. Tooth tips, magnet corners and slot openings also receive higher number of meshes.

4.2 No-Load Flux Density Distribution

Figure 8 shows the magnetic flux density on the full machine cross-section at no-load (zero phase current; magnets are the only source).

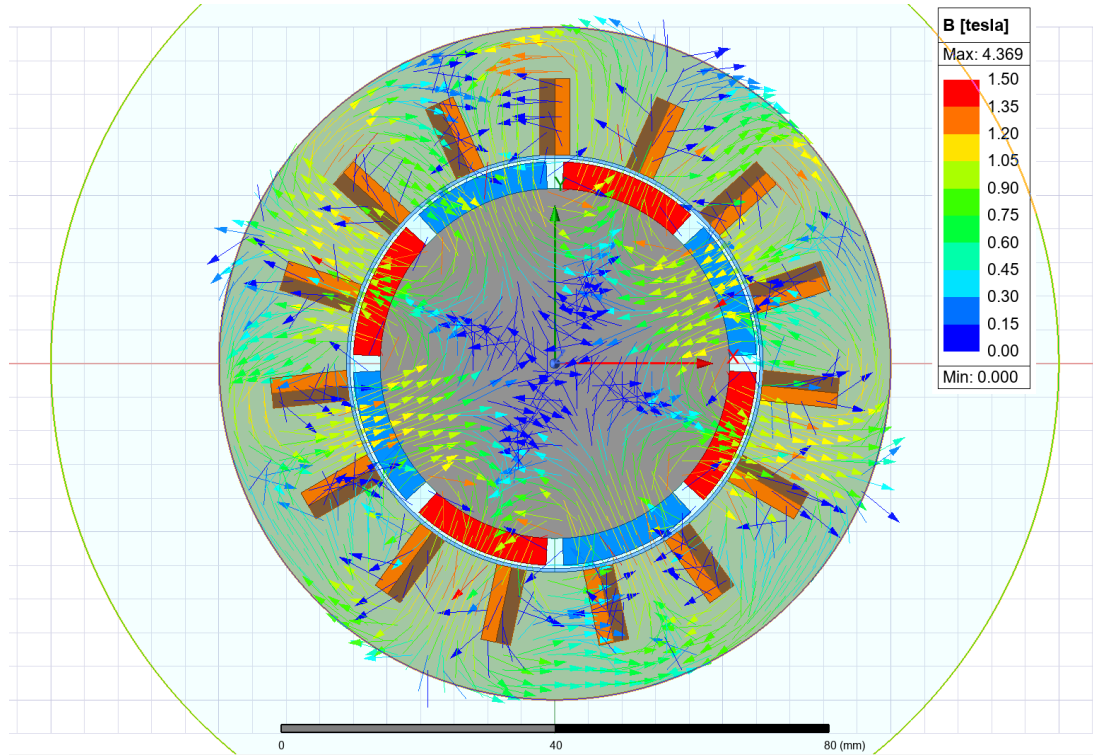


Figure 8: No-load flux density vectors over the full cross-section.

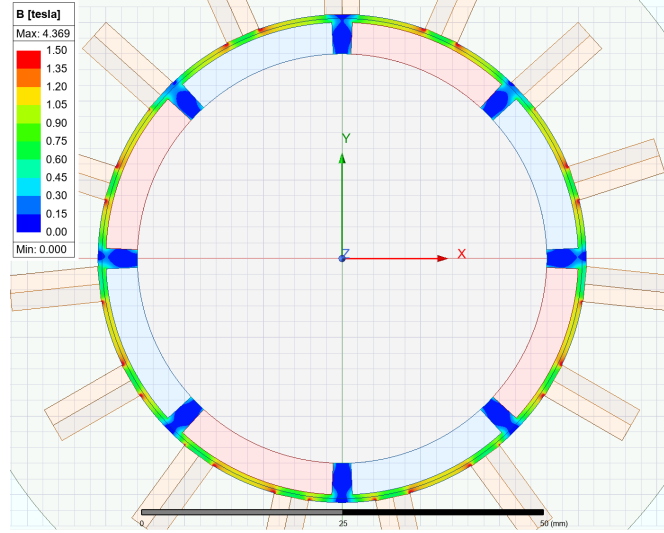
Figure 9: $|\mathbf{B}|$ across the air-gap.

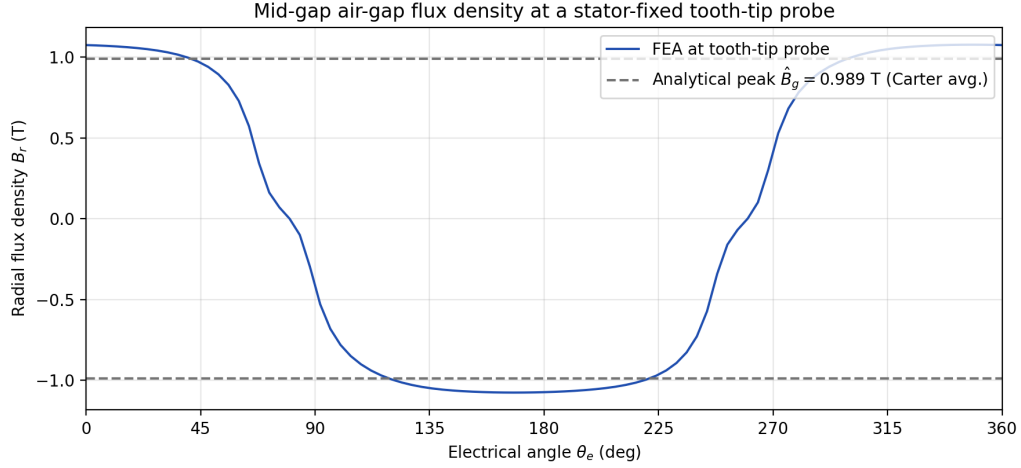
Table 16: Peak iron flux density – analytical vs. FEA (approximate values read from Figure 8).

Region	Analytical (Q3)	FEA (approx.)	Comment
Tooth body / tip	1.34 T	~ 1.3 T	saturated at legend cap
Stator yoke	1.40 T	~ 1.3 T	below analytical (under-utilised iron)
Rotor yoke	1.40 T	~ 1.3 T	similar
Magnet body	—	1.35–1.50 T	internal B near B_r , expected

The yokes sit slightly below the analytical target – a safe design margin – while the tooth tips push into the non-linear region locally, as expected for the geometrically narrower tooth-tip cross-section.

4.3 Air-Gap Flux Density

Because the 15/8 combination has no rotational symmetry ($\gcd(15, 8) = 1$), a single bore snapshot would mix the magnet pattern with the stator slot-permeance pattern. A cleaner approach is to fix a stator-frame probe at a tooth-tip centre and sweep the rotor over one electrical period, so each magnet passes by the same local airgap geometry. The probe sits at $r = 30.59$ mm, lab angle $\alpha_p = 154.44^\circ$, and the rotor sweeps $\theta_m = 0$ – 90° mechanical in 1° steps. At each rotor position B_x, B_y are exported and combined into $B_r = B_x \cos \alpha_p + B_y \sin \alpha_p$.

Figure 10: Mid-gap B_r at the tooth-tip probe vs. electrical angle.

The flat-top amplitude of the FEA waveform is $\hat{B}_{g,\text{FEA}} = 1.077 \text{ T}$, which exceeds the analytical peak by +8.9%. This is the expected behaviour at a tooth-tip probe: Carter's coefficient in the analytical model spatially averages the airgap permeance across one slot pitch.

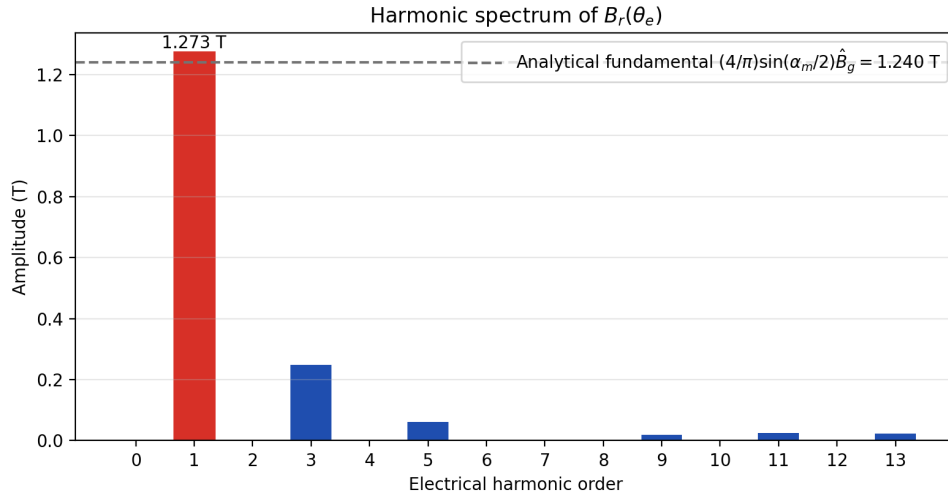
Figure 11: Harmonic spectrum of $B_r(\theta_e)$.

Table 17 summarises both metrics: the waveform peak (a tooth-centre value) and the fundamental Fourier component (the integrated quantity seen by the winding) – the latter agrees with the analytical $(4/\pi) \sin(\alpha_m/2) \hat{B}_g$ within 3%.

Table 17: Air-gap flux density: analytical vs. FEA.

Quantity	Analytical (Q3)	FEA (Q4)	Diff. (%)
Waveform peak $\hat{B}_g$	0.989 T	1.077 T	+8.9%
Fundamental of $B_r(\theta_e)$	1.240 T	1.273 T	+2.7%

4.4 Leakage Flux Paths

Three distinct leakage mechanisms are visible in zoomed views of the no-load field solution: magnet-to-magnet leakage between adjacent poles, in-slot leakage threading through the slot conductor region, and tooth-tip leakage hopping across a slot opening between adjacent tooth shoes.

Magnet-to-magnet leakage (Figure 12.) A small portion of the magnet flux short-circuits between adjacent N and S magnets through the inter-magnet air rather than crossing into the stator iron. It is a magnet-permeance loss that contributes nothing to torque production.

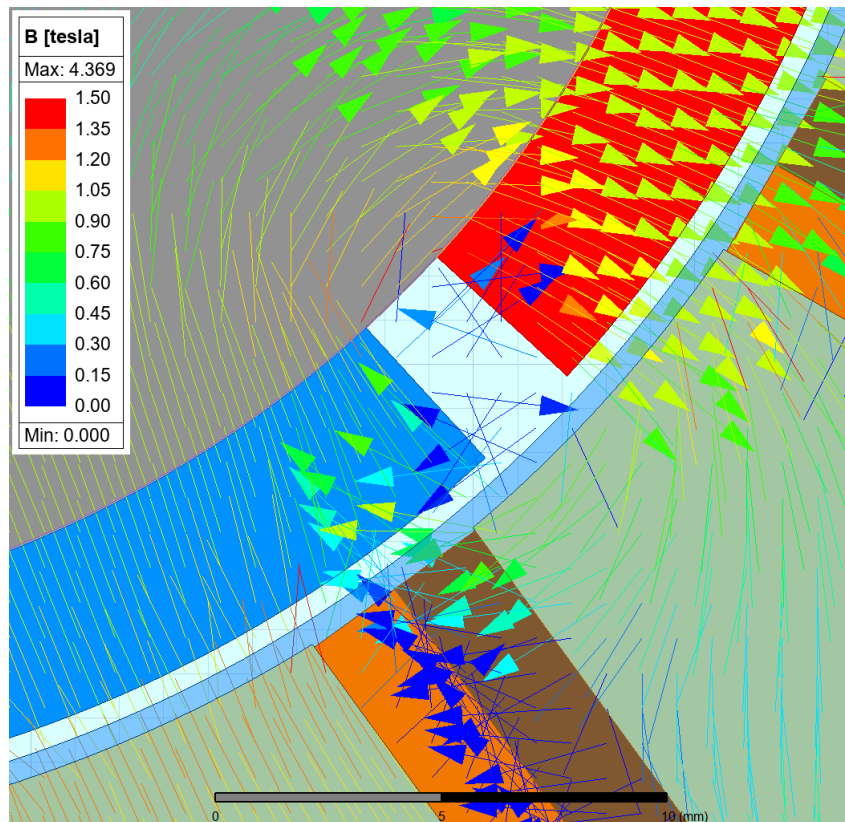


Figure 12: Magnet-to-magnet leakage.

In-slot leakage (Figure 13.) Flux looping inside the slot, threading from one tooth flank to the other through the conductor region. It contributes no useful torque-producing flux but stores reactive energy, appearing in lumped models as the slot-leakage inductance that raises the per-phase synchronous reactance and limits the machine's overload capability.

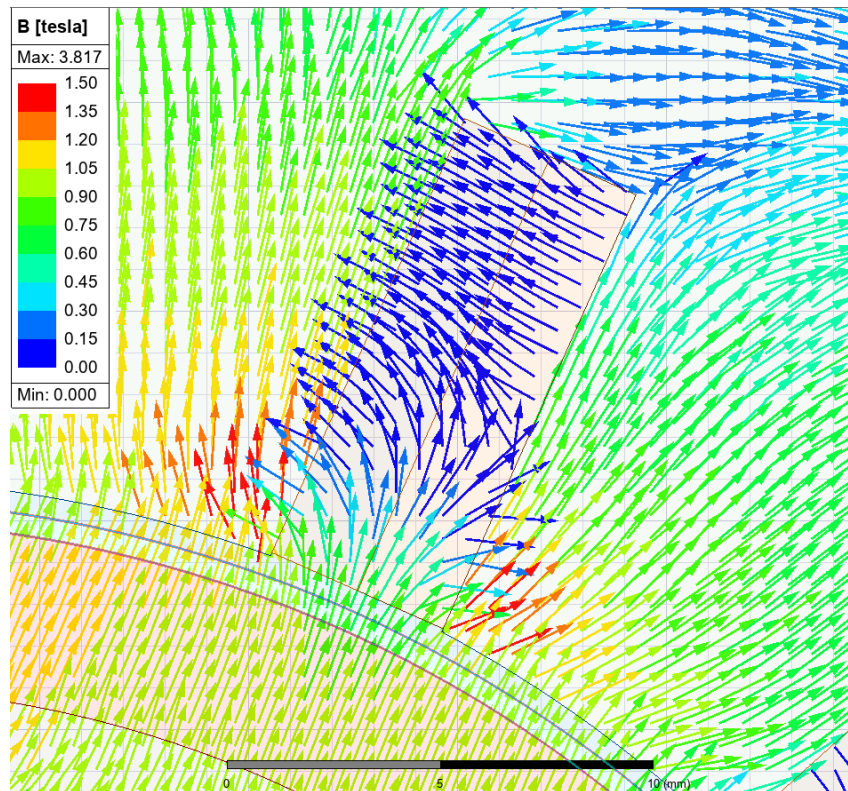


Figure 13: In-slot leakage.

Tooth-tip leakage (Figure 14.) A small fraction of the magnet flux hops across the slot opening between adjacent tooth-tip shoes rather than passing down the full tooth body to the yoke. This is the component Carter's coefficient parametrises in the analytical model.

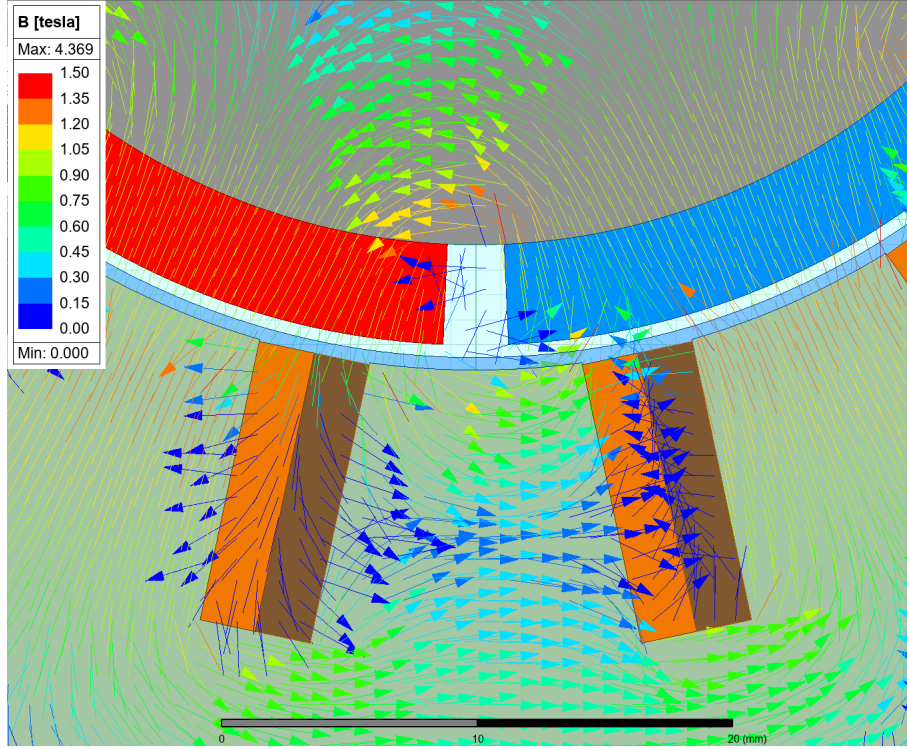


Figure 14: Tooth-tip leakage.

The magnet-to-magnet and tooth-tip paths together divert a small fraction of the magnet flux away from the useful air-gap, accounting for the $\sim 3\%$ under-prediction of the FEA flux per pole (§4.5) relative to the analytical 1D MEC.

4.5 Flux per Pole

By rotational invariance of the no-load problem, the temporal B_r record from §4.3 is equivalent to a spatial snapshot along 90° mechanical (= 2 pole pitches) of stator arc. The flux per pole follows by spatial integration over the mid-gap arc and multiplication by the effective stack length:

$$\Phi_{pole} = \frac{1}{2} L_e R_g \int_0^{\pi/2} |B_r(\theta_m)| d\theta_m \quad (80)$$

with $R_g = 30.5$ mm, $L_e = 102$ mm and the $1/2$ converting the two-pole integral to one pole.

Table 18: Flux per pole – analytical vs. FEA.

Source	Φ_{pole} (mWb)	Diff. vs. analytical
Analytical (Q3)	2.18	—
FEA, mid-gap integration	2.11	-3.0%

The spatial integral averages high-permeance tooth regions against low-permeance slot openings – exactly the Carter average embedded in the analytical model – and the $\sim 3\%$ residual is attributable to leakage that the 1D MEC ignores.

4.6 Cogging Torque (Bonus)

Cogging torque is extracted from the same parametric sweep used in §4.3: with all phase currents at zero, Maxwell evaluates the rotor torque by the Virtual Work method at each rotor position. The sweep range 0–90° mechanical contains $90/3 = 30$ cogging cycles, since the 15/8 combination has cogging period $\theta_{\text{cog}} = 360^\circ/\text{lcm}(15, 8) = 3^\circ$ mechanical.

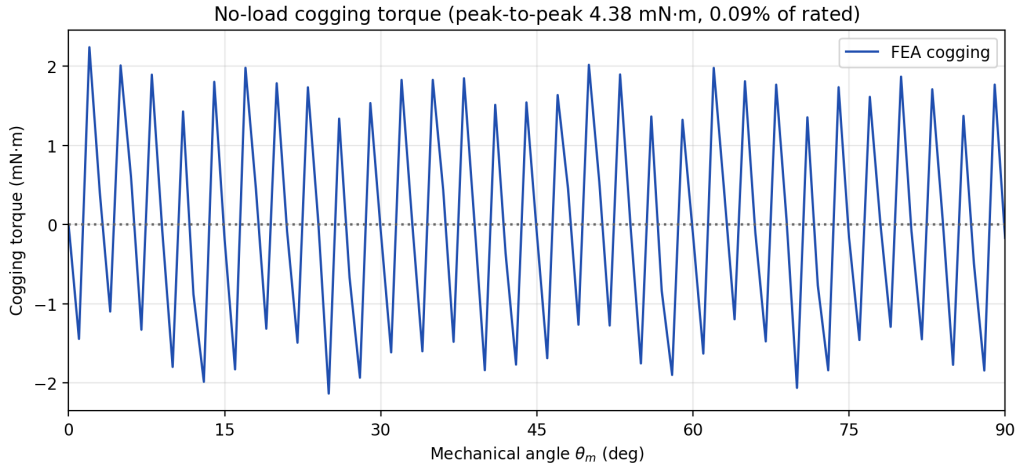


Figure 15: No-load cogging torque vs. rotor angle.

Peak-to-peak amplitude is $T_{\text{cog,pp}} = 4.38 \text{ mN}\cdot\text{m}$ (RMS $1.39 \text{ mN}\cdot\text{m}$), i.e. 0.090% of the rated torque $4.85 \text{ N}\cdot\text{m}$ – a very low cogging level, characteristic of FSCW designs with $\text{lcm}(15, 8) = 120$.

4.7 Summary – Analytical vs. FEA

Table 19: Final comparison of analytical and FEA results for Design 1.

Quantity	Analytical (Q3)	FEA (Q4)	Diff. (%)
Air-gap peak $\hat{B}_g$ (waveform top)	0.989 T	1.077 T	+8.9
Air-gap fundamental of B_r	1.240 T	1.273 T	+2.7
Flux per pole Φ_{pole}	2.18 mWb	2.11 mWb	−3.0
Tooth flux density B_{tooth}	1.34 T	~1.3 T	—
Yoke flux density B_{yoke}	1.40 T	~1.3 T	—
Cogging T_{p-p}	—	4.4 mN·m (0.09%)	—

Agreement between the analytical and the 2D FEA is similar for every quantity, which is expected because Carter’s coefficient in the analytical model spatially averages over the slot pitch and so under-predicts the local tooth-tip value.

A MATLAB Script – Analytical Calculations (Q1–Q3)

The full MATLAB script `ee568_hw2_analytical.m` is reproduced below. It computes all Q1–Q3 results, generates the winding diagrams and phasor plots in Section 2 (Figures 1–3), and prints a summary table to the console.

```

1 %% EE568 HW2 - Winding Design & Analytical Modelling
2 % Student : Sefer İDACI | ID: 2575421
3 % Design : Design 1 (ID mod 5 = 1) → Ns = 15, Nm = 8
4 % Deadline: 01 June 2026
5
6 clear; clc; close all;
7
8 %% =====
9 %% Q1 - INTEGRAL-SLOT WINDING (6-pole, 72-slot, 3-phase DL)
10 %% =====
11 fprintf('=== Q1: Integral-Slot Winding (72-slot, 6-pole, 3-phase) ===\n\n');
12
13 Ns1 = 72; p1 = 3; m = 3; % slots, pole-pairs, phases
14 ae1 = p1*360/Ns1; % elec. deg per slot = 15°
15 q1 = Ns1/(2*p1*m); % slots/pole/phase = 4
16 tau1 = Ns1/(2*p1); % pole pitch in slots = 12
17
18 fprintf(' Slot pitch (elec.) ae = %.1f°\n', ae1);
19 fprintf(' Slots/pole/phase q = %.4g\n', q1);
20 fprintf(' Pole pitch tau = %d slots\n\n', tau1);
21
22 configs = [tau1, 11]; % full-pitch, 11/12 short-pitch
23 cnames = {'Full-pitched (y=12)', '11/12 Short-pitched (y=11)'};
24 harmonics = [1, 3, 5];
25
26 for ci = 1:2
27     y = configs(ci);
28     fprintf('--- %s ---\n', cnames{ci});
29     fprintf(' %-22s %8s %8s %8s\n', 'Harmonic', 'kd', 'kp', 'kw');
30     fprintf(' %s\n', repmat('-',1,50));
31     for n = harmonics
32         kd = sin(n*q1*ae1/2 * pi/180) / (q1 * sin(n*ae1/2 * pi/180));
33         kp = sin(n*y*pi / (2*tau1));
34         kw = abs(kd) * abs(kp);
35         fprintf(' %-22s %8.4f %8.4f %8.4f\n', sprintf('n = %d',n), abs(kd), abs(kp), kw);
36     end
37     fprintf('\n');
38 end
39
40 % ---- Q1 winding diagram (one pole-pair = 24 slots, slots 1..24) ----
41 q1_slots = 2*p1*m*q1; % slots per pole-pair = 24 (one period)
42 slots_pp = 1 : q1_slots;
43
44 % Assign upper layer: sectors of 60° each, A+(0-60) C-(60-120) B+(120-180)
45 % A-(180-240) C+(240-300) B-(300-360)
46 ang1 = mod((slots_pp-1)*ae1, 360);
47 [phase1_up, dir1_up] = assign_phase(ang1); % helper at end of file
48
49 % Lower layer is shifted by (y) slots (coil return path)
50 ang1_lo = mod((slots_pp-1+configs(1))*ae1, 360); % full-pitch lower
51 [ph1_lo_fp, di1_lo_fp] = assign_phase(ang1_lo);
52 ang1_lo_sp = mod((slots_pp-1+configs(2))*ae1, 360); % short-pitch lower
53 [ph1_lo_sp, di1_lo_sp] = assign_phase(ang1_lo_sp);
54
55 figure('Name','Q1 Winding Layout - One Pole-Pair','Position',[50 50 1100 500]);
56 for ci = 1:2
57     subplot(1,2,ci);
58     if ci==1; ph_lo=ph1_lo_fp; di_lo=di1_lo_fp; else; ph_lo=ph1_lo_sp; di_lo=di1_lo_sp; end
59     draw_winding_layout(slots_pp, phase1_up, dir1_up, ph_lo, di_lo, cnames{ci});
60 end
61 saveas(gcf, 'q1_winding_layout.png');
62
63 %% =====
64 %% Q2 - FRACTIONAL-SLOT WINDING (15-slot, 8-pole, 3-phase)
65 %% =====

```

```

66 fprintf('=== Q2: Fractional-Slot Winding (15-slot, 8-pole, 3-phase) ===\n\n');
67
68 Ns = 15; Nm = 8; p = Nm/2; m = 3;
69 as = Nm*180/Ns; % electrical degrees per slot = 96°
70 q = Ns/(Nm*m); % slots per pole per phase = 0.625
71 sa = mod((0:Ns-1)*as, 360); % slot phasor angles [°], slots indexed 1..Ns
72
73 fprintf(' Electrical angle per slot as = %.2f\n', as);
74 fprintf(' Slots per pole per phase q = %.4f\n\n', q);
75
76 % Slot angle table
77 fprintf(' Slot | Angle (°)\n');
78 fprintf(' -----\n');
79 for k = 1:Ns
80     fprintf(' %2d | %6.1f\n', k, sa(k));
81 end
82
83 % =====
84 % USER INPUT -- edit these 6 arrays to change the winding assignment
85 %
86 % Each array lists 5 slot numbers for one phase + one layer.
87 % Layer 1: first n_pos_L1 entries = positive slots (current IN, +)
88 %           remaining entries = negative slots (current OUT, -)
89 % Layer 2: first n_pos_L2 entries = positive slots
90 %           remaining entries = negative slots
91 %
92 % Slot angles: sa(k) = (k-1)*96° mod 360°
93 % k=1: 0°, k=2: 96°, k=3: 192°, k=4: 288°, k=5: 24°, k=6: 120°,
94 % k=7: 216°, k=8: 312°, k=9: 48°, k=10: 144°, k=11: 240°,
95 % k=12: 336°, k=13: 72°, k=14: 168°, k=15: 264°
96 % =====
97 n_pos_L1 = 3; % Layer 1: first 3 entries are positive (+), last 2 are negative (-)
98 n_pos_L2 = 2; % Layer 2: first 2 entries are positive (+), last 3 are negative (-)
99
100 % Winding (kw=0.9566)
101 % [ + + + - - ]
102 A_layer1 = [ 1, 5, 9, 3, 7 ];
103 B_layer1 = [ 6, 10, 14, 8, 12 ];
104 C_layer1 = [ 4, 11, 15, 2, 13 ];
105
106 % [ + + - - - ]
107 A_layer2 = [ 4, 8, 2, 6, 10 ];
108 B_layer2 = [ 9, 13, 7, 11, 15 ];
109 C_layer2 = [ 3, 14, 1, 5, 12 ];
110
111
112
113 % =====
114 % Winding factor -- phasor summation for n = 1, 3, 5
115 % =====
116 phase_labels = {'A', 'B', 'C'};
117 L1_all = {A_layer1; B_layer1; C_layer1};
118 L2_all = {A_layer2; B_layer2; C_layer2};
119
120 kw_L1 = zeros(3,3); % rows = phase (A,B,C), cols = harmonic (n=1,3,5)
121
122 for ph = 1:3
123     L1 = L1_all{ph};
124     pos_L1 = L1(1:n_pos_L1); % positive slots, layer 1
125     neg_L1 = L1(n_pos_L1+1:end); % negative slots, layer 1
126
127     for hi = 1:3
128         n = harmonics(hi);
129         % positive slots: phasor at n*angle
130         % negative slots: phasor at n*angle + 180° (current reversal)
131         ph_L1 = [n*sa(pos_L1), n*sa(neg_L1)+180];
132         kw_L1(ph,hi) = abs(sum(exp(1j * ph_L1 * pi/180))) / 5;
133     end
134 end
135
136 kw_q2 = kw_L1(1,:); % Phase A winding factors -- used in Q3
137
138 fprintf('\n Winding factors (concentrated-coil convention):\n');

```

```

139 fprintf(' Phase |      n=1      n=3      n=5\n');
140 fprintf(' %s\n', repmat('-',1,35));
141 for ph = 1:3
142     fprintf('      %s |      %.4f      %.4f      %.4f\n', phase_labels{ph}, kw_L1(ph,:));
143 end
144
145 % =====
146 % Build per-slot phase/direction from Layer 1 (used by layout plot)
147 % ph2 values: 1=A+, 2=A-, 3=B+, 4=B-, 5=C+, 6=C-
148 % =====
149 ph2 = zeros(1,Ns);
150 dir2 = zeros(1,Ns);
151 for ph = 1:3
152     L1 = L1_all{ph};
153     ph2(L1(1:n_pos_L1)) = 2*ph-1; dir2(L1(1:n_pos_L1)) = +1;
154     ph2(L1(n_pos_L1+1:end)) = 2*ph; dir2(L1(n_pos_L1+1:end)) = -1;
155 end
156
157 % Build per-slot phase/direction from Layer 2 (for Layer 2 star plot)
158 ph2_L2 = zeros(1,Ns);
159 dir2_L2 = zeros(1,Ns);
160 for ph = 1:3
161     L2 = L2_all{ph};
162     ph2_L2(L2(1:n_pos_L2)) = 2*ph-1; dir2_L2(L2(1:n_pos_L2)) = +1;
163     ph2_L2(L2(n_pos_L2+1:end)) = 2*ph; dir2_L2(L2(n_pos_L2+1:end)) = -1;
164 end
165
166 % =====
167 % Shared plot settings (change once, applies to all figures)
168 % =====
169 cA = [0.85 0.15 0.15]; % Phase A colour
170 cB = [0.15 0.45 0.85]; % Phase B colour
171 cC = [0.15 0.70 0.20]; % Phase C colour
172 cArrow = 'b'; % phasor arrow colour
173 cResultant = 'r'; % resultant arrow colour
174 arrowWidth = 1.2; % individual phasor line width
175 resultWidth = 2.5; % resultant arrow line width
176 theta_circle = linspace(0, 2*pi, 200); % unit circle
177
178 % =====
179 % Figure 1 -- Star of slots (each slot coloured by phase)
180 % =====
181 phase_colors = {cA, cA, cB, cB, cC, cC}; % index 1..6 maps to ph2
182
183 figure('Name','Q2 Star of Slots','Position',[50 600 560 480]);
184 hold on;
185 for k = 1:Ns
186     ang = sa(k) * pi/180;
187     c = phase_colors{ph2(k)};
188     quiver(0,0, cos(ang), sin(ang), 0, ...
189         'Color',c, 'LineWidth',1.5, 'MaxHeadSize',0.3);
190     text(1.15*cos(ang), 1.15*sin(ang), num2str(k), ...
191         'FontSize',8, 'HorizontalAlignment','center', 'Color',c);
192 end
193 plot(cos(theta_circle), sin(theta_circle), 'k--', 'LineWidth',0.5);
194 axis equal; xlim([-1.4 1.4]); ylim([-1.4 1.4]); grid on;
195 xlabel('Real'); ylabel('Imaginary');
196 title('Star of Slots - 15-slot / 8-pole');
197 legend([patch(NaN,NaN,cA), patch(NaN,NaN,cB), patch(NaN,NaN,cC)], ...
198     'Phase A','Phase B','Phase C', 'Location','southeast');
199 saveas(gcf, 'q2_star_of_slots.png');
200
201 % =====
202 % Figure 2 -- Phase A phasors (n = 1, 3, 5)
203 % =====
204 pos_A = A_layer1(1:n_pos_L1);
205 neg_A = A_layer1(n_pos_L1+1:end);
206
207 figure('Name','Q2 Phase A Phasors','Position',[630 600 900 300]);
208 for hi = 1:3
209     n = harmonics(hi);
210     all_ph = [n*sa(pos_A), n*sa(neg_A)+180];
211     psum = sum(exp(1j * all_ph * pi/180));

```

```

212     subplot(1,3,hi); hold on;
213     for ii = 1:length(all_ph)
214         a = all_ph(ii)*pi/180;
215         quiver(0,0, cos(a),sin(a), 0, cArrow, 'LineWidth',arrowWidth, 'MaxHeadSize',0.4);
216     end
217     quiver(0,0, real(psum)/5, imag(psum)/5, 0, ...
218         cResultant, 'LineWidth',resultWidth, 'MaxHeadSize',0.3);
219     plot(cos(theta_circle), sin(theta_circle), 'k--', 'LineWidth',0.5);
220     axis equal; xlim([-1.4 1.4]); ylim([-1.4 1.4]); grid on;
221     title(sprintf('Phase A, n=%d kw = %.4f', n, kw_L1(1,hi)));
222     xlabel('Real'); ylabel('Imaginary');
223 end
224 saveas(gcf, 'q2_phase_A_phasors.png');
225
226 % =====
227 % Figure 3 -- Stator cross-section with teeth and double-layer slots
228 % =====
229 figure('Name','Q2 Stator Winding Layout','Position',[50 100 800 800]);
230 draw_stator_winding(Ns, ph2, dir2, ph2_L2, dir2_L2, ...
231     'Design - 15-slot/8-pole, Double-Layer FSCW');
232 saveas(gcf, 'q2_winding_layout.png');
233
234 %% =====
235 %% Q3 - ANALYTICAL MODELLING
236 %% =====
237 fprintf('\n=== Q3: Analytical Modelling ===\n\n');
238
239 %--- Common specs -----
240 Rso = 50e-3; % stator outer radius [m]
241 Rro = 0.6*Rso; % rotor outer radius [m] = 30 mm
242 g = 1e-3; % air-gap length [m]
243 lm = 4e-3; % magnet radial thickness [m]
244 Br = 1.3; % remanence [T]
245 mur = 1.05; % magnet relative permeability
246 am = 160/180; % magnet arc ratio (160° elec.)
247 Ls = 100e-3; % stack length [m]
248 J = 5e6; % current density (rms) [A/m²]
249 kwc = 0.60; % conductor fill factor
250 Bt = 1.4; % target peak flux density in iron [T]
251 Vdc = 48; % DC bus voltage [V]
252 n_rpm = 1500; % rated speed [rpm]
253 kw1 = kw_q2(1); % winding factor (fundamental)
254
255 %--- Derived radii -----
256 Rsi = Rro + g; % stator inner radius [m] = 31 mm
257 Rro_core = Rro - lm; % rotor core outer radius [m] = 26 mm
258
259 fprintf('Machine Geometry\n');
260 fprintf(' Rso (stator outer) = %.1f mm\n', Rso*1e3);
261 fprintf(' Rsi (stator inner) = %.1f mm\n', Rsi*1e3);
262 fprintf(' Rro (rotor outer) = %.1f mm\n', Rro*1e3);
263 fprintf(' Rro_core (core outer) = %.1f mm\n', Rro_core*1e3);
264 fprintf(' Air gap g = %.1f mm\n', g*1e3);
265 fprintf(' Magnet thickness lm = %.1f mm\n', lm*1e3);
266
267 %--- Slot pitch at stator bore -----
268 tau_s = 2*pi*Rsi / Ns; % slot pitch [m]
269 fprintf('Stator Slot Geometry\n');
270 fprintf(' Slot pitch at bore  $\tau_s$  = %.4f mm\n', tau_s*1e3);
271
272 %--- First-pass air-gap flux density (no Carter) -----
273 Bg0 = Br*lm / (lm + mur*g);
274 fprintf('\n Bg_peak (no Carter) = %.4f T\n', Bg0);
275
276 %--- Tooth width for Bt = 1.4 T -----
277 Bg_avg0 = Bg0 * am;
278 wtb = Bg_avg0 * tau_s / Bt;
279 ws = tau_s - wtb;
280 fprintf(' Tooth body width wtb = %.4f mm\n', wtb*1e3);
281 fprintf(' Slot opening width ws = %.4f mm\n', ws*1e3);
282
283 %--- Carter's coefficient (exact formula) -----

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285 x = ws/(2*g);
286 gam = (4/pi) * (x*atan(x) - log(sqrt(1+x^2))); % length [m] when multiplied by g
287 kc = tau_s / (tau_s - gam*g);
288 fprintf('\nCarter's Coefficient\n');
289 fprintf(' ws / g = %.3f\n', ws/g);
290 fprintf(' γ (slot fringe) = %.4f mm\n', gam*g*1e3);
291 fprintf(' Carter coefficient kc = %.4f\n', kc);
292
293 %--- Effective air gap & corrected Bg -----
294 g_eff = kc*g + lm/mur;
295 Bg = Br*lm / (lm + mur*kc*g);
296 fprintf(' g_eff = kc*g + lm/μr = %.4f mm\n', g_eff*1e3);
297 fprintf(' Bg_peak (corrected) = %.4f T\n', Bg);
298
299 %--- Effective axial length (fringing correction) -----
300 Le = Ls + 2*g;
301 fprintf('\nEffective Axial Length\n');
302 fprintf(' Ls (physical) = %.1f mm\n', Ls*1e3);
303 fprintf(' Le = Ls + 2g = %.2f mm\n', Le*1e3);
304
305 %--- Pole pitch & flux per pole -----
306 tau_p = pi*Rsi / p; % pole pitch at stator bore [m]
307 Phi = Bg * am * tau_p * Le; % flux per pole [Wb]
308 fprintf('\nFlux Per Pole\n');
309 fprintf(' Pole pitch τp = %.4f mm\n', tau_p*1e3);
310 fprintf(' Flux per pole Φ = %.6f Wb (%.4f mWb)\n', Phi, Phi*1e3);
311
312 %--- Stator and rotor yoke widths -----
313 wsy = Phi / (2 * Bt * Le);
314 wry = Phi / (2 * Bt * Le);
315 fprintf('\nIron Yoke Widths\n');
316 fprintf(' Stator yoke wsy = %.4f mm\n', wsy*1e3);
317 fprintf(' Rotor yoke wry = %.4f mm\n', wry*1e3);
318 fprintf(' Rotor shaft radius = %.4f mm\n', (Rro_core - wry)*1e3);
319
320 %--- Slot dimensions & electrical loading -----
321 h_total = Rso - Rsi;
322 h_slot = h_total - wsy;
323 A_slot = ws * h_slot; % slot area (simplified rectangular) [m²]
324 A_rms = J * kwc * A_slot / tau_s; % linear current density rms [A/m]
325 A_peak = A_rms * sqrt(2);
326
327 fprintf('\nElectrical Loading\n');
328 fprintf(' Stator radial height = %.4f mm\n', h_total*1e3);
329 fprintf(' Slot height h_slot = %.4f mm\n', h_slot*1e3);
330 fprintf(' Slot area (rect.) = %.4f mm²\n', A_slot*1e6);
331 fprintf(' A_rms (linear) = %.2f kA/m\n', A_rms/1e3);
332
333 %--- Magnetic loading (average Bg over rotor surface) -----
334 Bav = Bg * am;
335 fprintf('\nMagnetic Loading\n');
336 fprintf(' Bav = Bg·am = %.4f T\n', Bav);
337
338 %--- Specific machine constant & shear stress -----
339 % Average electromagnetic shear stress: σ = kw·Bav·A_rms (N/m²)
340 % More precisely: σ = (1/2)·kw·Bg_peak·A_peak
341 sigma = 0.5 * kw1 * Bg * A_peak;
342 fprintf('\nShear Stress\n');
343 fprintf(' σ = ½·kw·Bg·A_peak = %.2f kPa\n', sigma/1e3);
344
345 %--- Expected torque & power -----
346 T_exp = sigma * 2*pi * Rsi^2 * Le;
347 omega = n_rpm * 2*pi / 60;
348 P_exp = T_exp * omega;
349 fprintf('\nPerformance Estimates\n');
350 fprintf(' Expected torque = %.3f N·m\n', T_exp);
351 fprintf(' Rated speed ω = %.2f rad/s\n', omega);
352 fprintf(' Expected power = %.1f W\n', P_exp);
353
354 %--- Number of turns per phase (for Vdc=48V, n=1500rpm) -----
355 omega_e = p * omega;
356 Vph_peak = Vdc / sqrt(3); % peak phase voltage (star, full modulation)
357 Nph_float = Vph_peak / (kw1 * omega_e * Phi);

```

```

358 Ncs_float = 2*Nph_float*m / Ns; % conductors per slot (must be integer)
359 % Round to nearest integer conductors/slot
360 Ncs = round(Ncs_float);
361 Nph = Ncs * Ns / (2*m);
362 Vph_actual = kw1 * Nph * omega_e * Phi;
363
364 fprintf('\nWinding Turns (for Vdc = %.0f V, n = %.0f rpm)\n', Vdc, n_rpm);
365 fprintf('  we (elec. freq.)      = %.2f rad/s\n', omega_e);
366 fprintf('  Vph_peak (star)         = %.2f V\n', Vph_peak);
367 fprintf('  Nph (ideal)              = %.2f turns/phase\n', Nph_float);
368 fprintf('  Ncs (conductors/slot)    = %d (rounded to integer)\n', Ncs);
369 fprintf('  Nph (actual)            = %d turns/phase\n', Nph);
370 fprintf('  Back-EMF (peak)         = %.2f V\n', Vph_actual);
371
372 %--- Summary table for report -----
373 fprintf('\n=== Summary Table ===\n');
374 fprintf('  %-35s %12s\n', 'Parameter', 'Value');
375 fprintf('  %s\n', repmat('-',1,50));
376 params = {
377     'Stator inner radius Rsi',      sprintf('%.1f mm', Rsi*1e3);
378     'Effective air gap g_eff',      sprintf('%.3f mm', g_eff*1e3);
379     'Carter coefficient kc',        sprintf('%.4f', kc);
380     'Effective axial length Le',    sprintf('%.1f mm', Le*1e3);
381     'Air-gap peak flux density Bg', sprintf('%.4f T', Bg);
382     'Flux per pole Φ',              sprintf('%.4f mWb', Phi*1e3);
383     'Average flux density Bav',     sprintf('%.4f T', Bav);
384     'Electrical loading A_rms',     sprintf('%.2f kA/m', A_rms/1e3);
385     'Shear stress σ',              sprintf('%.2f kPa', sigma/1e3);
386     'Expected torque',             sprintf('%.3f N-m', T_exp);
387     'Expected power',             sprintf('%.1f W', P_exp);
388     'Turns per phase Nph',         sprintf('%d', Nph);
389 };
390 for i = 1:size(params,1)
391     fprintf('  %-35s %12s\n', params{i,1}, params{i,2});
392 end
393
394 fprintf('\nDone. Figures saved to HW2 folder.\n');
395
396 %% =====
397 %% LOCAL HELPER FUNCTIONS
398 %% =====
399
400 function [ph, d] = assign_phase(angles)
401 % Returns phase index (1=A+, 2=A-, 3=B+, 4=B-, 5=C+, 6=C-)
402 % and direction (+1 or -1) for each angle in [0,360).
403 n = length(angles);
404 ph = zeros(1,n);
405 d = zeros(1,n);
406 for k = 1:n
407     a = mod(angles(k), 360);
408     if a < 60, ph(k)=1; d(k)= 1; % A+
409     elseif a < 120, ph(k)=6; d(k)=-1; % C- (ph=6 for C-, reuse index)
410     elseif a < 180, ph(k)=3; d(k)= 1; % B+
411     elseif a < 240, ph(k)=2; d(k)=-1; % A-
412     elseif a < 300, ph(k)=5; d(k)= 1; % C+
413     else, ph(k)=4; d(k)=-1; % B-
414     end
415 end
416 % Remap to simple phase 1=A, 2=A, 3=B, 4=B, 5=C, 6=C
417 % (direction stored separately in d)
418 end
419
420 function draw_winding_layout(slots, ph_up, dir_up, ph_lo, dir_lo, ttl)
421 % Simple bar chart layout for one pole-pair of integral-slot winding.
422 cmap = [0.85 0.15 0.15; % A+ red
423         0.15 0.45 0.85; % C- blue
424         0.15 0.70 0.20; % B+ green
425         0.85 0.15 0.15; % A- red
426         0.15 0.70 0.20; % C+ green
427         0.15 0.45 0.85]; % B- blue
428 Ns = length(slots);
429 hold on;
430 for k = 1:Ns

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431     % upper layer (y = 0.5 .. 1)
432     fill([k-0.9 k-0.1 k-0.1 k-0.9], [0.5 0.5 1 1], cmap(ph_up(k,:),:), ...
433         'EdgeColor','k','LineWidth',0.5);
434     % lower layer (y = 0 .. 0.5)
435     fill([k-0.9 k-0.1 k-0.1 k-0.9], [0 0 0.5 0.5], cmap(ph_lo(k,:),:), ...
436         'EdgeColor','k','LineWidth',0.5,'FaceAlpha',0.5);
437     % direction arrows
438     if dir_up(k) > 0; sym = '\bullet'; else; sym = '\times'; end
439     text(k-0.5, 0.75, sym, 'HorizontalAlignment','center','FontSize',7);
440     if dir_lo(k) > 0; sym = '\bullet'; else; sym = '\times'; end
441     text(k-0.5, 0.25, sym, 'HorizontalAlignment','center','FontSize',7);
442 end
443 xlim([0 Ns+1]); ylim([-0.2 1.3]); axis off;
444 title(ttl, 'FontSize', 9);
445 legend([patch(NaN,NaN,[0.85 0.15 0.15]), ...
446         patch(NaN,NaN,[0.15 0.45 0.85]), ...
447         patch(NaN,NaN,[0.15 0.70 0.20])], ...
448         'Phase A','Phase B','Phase C','Location','south','FontSize',7);
449 end
450
451 function draw_winding_layout_radial(Ns, ph2, dir2, ttl)
452 % Circular slot layout for fractional-slot winding.
453 cA = [0.85 0.15 0.15];
454 cB = [0.15 0.45 0.85];
455 cC = [0.15 0.70 0.20];
456 cmap_full = {cA, cA, cB, cB, cC, cC}; % index 1..6
457
458 hold on; axis equal; axis off;
459 R = 1.0; w = 0.25;
460 angles_slot = linspace(90, 90-360, Ns+1) * pi/180;
461 for k = 1:Ns
462     th1 = angles_slot(k);
463     th2 = angles_slot(k+1);
464     th = linspace(th1, th2, 20);
465     % outer arc polygon
466     xo = R*cos(th); yo = R*sin(th);
467     xi = (R-w)*cos(fliplr(th)); yi = (R-w)*sin(fliplr(th));
468     c = cmap_full{ph2(k)};
469     fill([xo xi], [yo yi], c, 'EdgeColor','k','LineWidth',0.6);
470     % direction symbol
471     tmid = (th1+th2)/2;
472     rm = R - w/2;
473     if dir2(k) > 0; sym = '\bullet'; else; sym = '\times'; end
474     text(rm*cos(tmid), rm*sin(tmid), sym, ...
475         'HorizontalAlignment','center','FontSize',7,'Color','w','FontWeight','bold');
476     % slot number
477     text(1.15*cos(tmid), 1.15*sin(tmid), num2str(k), ...
478         'HorizontalAlignment','center','FontSize',7);
479 end
480 title(ttl, 'FontSize', 10);
481 legend([patch(NaN,NaN,cA), patch(NaN,NaN,cB), patch(NaN,NaN,cC)], ...
482         'Phase A','Phase B','Phase C','Location','south');
483 end
484
485 function draw_stator_winding(Ns, ph_L1, dir_L1, ph_L2, dir_L2, ttl)
486 % Stator cross-section for an FSCW (single-tooth concentrated winding).
487 % Each slot holds two coil sides SIDE-BY-SIDE (tangential split):
488 % - The half nearer tooth k = go-side of coil k (Layer 1)
489 % - The half nearer tooth k-1 = return-side of coil k-1 (Layer 2)
490 % Slots ordered clockwise with slot 1 at the top.
491
492 % Phase colours
493 cA = [0.95 0.55 0.55];
494 cB = [0.55 0.70 0.95];
495 cC = [0.55 0.85 0.60];
496 phase_colors = {cA, cA, cB, cB, cC, cC}; % index 1..6
497
498 % Geometry (normalised)
499 R_yoke_out = 1.00;
500 R_yoke_in = 0.85; % yoke inner edge / tooth root
501 R_bore = 0.55; % stator bore / tooth tip
502 R_rotor = 0.50;
503 tooth_frac = 0.4; % tooth angular width / slot pitch

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504
505 slot_pitch = 2*pi / Ns;
506 tooth_half = tooth_frac * slot_pitch / 2;
507 slot_half = (1-tooth_frac) * slot_pitch / 2;
508 theta0 = pi/2;
509
510 hold on; axis equal; axis off;
511 th_full = linspace(0, 2*pi, 360);
512
513 % --- Back-iron annulus -----
514 fill([R_yoke_out*cos(th_full), fliplr(R_yoke_in*cos(th_full))], ...
515      [R_yoke_out*sin(th_full), fliplr(R_yoke_in*sin(th_full))], ...
516      [0.80 0.80 0.80], 'EdgeColor','k','LineWidth',0.8);
517
518 % --- Teeth -----
519 for k = 1:Ns
520     slot_c = theta0 - (k-1)*slot_pitch;
521     tooth_c = slot_c - slot_pitch/2; % tooth k between slot k and slot k+1
522     th_t = linspace(tooth_c - tooth_half, tooth_c + tooth_half, 20);
523     xo = R_yoke_in*cos(th_t); yo = R_yoke_in*sin(th_t);
524     xi = R_bore *cos(fliplr(th_t)); yi = R_bore *sin(fliplr(th_t));
525     fill([xo xi], [yo yi], [0.75 0.75 0.75], ...
526          'EdgeColor','k','LineWidth',0.6);
527 end
528
529 % --- Slot fillings (TANGENTIAL split: L1 right side, L2 left side)
530 phase_letter = {'A','A','B','B','C','C'};
531 for k = 1:Ns
532     slot_c = theta0 - (k-1)*slot_pitch;
533
534     % Slot 1 is at theta = pi/2 (top); slots advance CLOCKWISE so
535     % "smaller theta" is to the RIGHT of the slot (closer to tooth k),
536     % "larger theta" is to the LEFT (closer to tooth k-1).
537     th_R = linspace(slot_c - slot_half, slot_c, 14); % right side (Layer 1)
538     th_L = linspace(slot_c, slot_c + slot_half, 14); % left side (Layer 2)
539
540     % --- Right side of slot = Layer 1 (go-side of coil k) -----
541     c1 = phase_colors{ph_L1(k)};
542     xo = R_yoke_in*cos(th_R); yo = R_yoke_in*sin(th_R);
543     xi = R_bore *cos(fliplr(th_R)); yi = R_bore *sin(fliplr(th_R));
544     fill([xo xi], [yo yi], c1, 'EdgeColor','k','LineWidth',0.4);
545
546     % --- Left side of slot = Layer 2 (return of coil k-1) -----
547     c2 = phase_colors{ph_L2(k)};
548     xo = R_yoke_in*cos(th_L); yo = R_yoke_in*sin(th_L);
549     xi = R_bore *cos(fliplr(th_L)); yi = R_bore *sin(fliplr(th_L));
550     fill([xo xi], [yo yi], c2, 'EdgeColor','k','LineWidth',0.4);
551
552     % --- Radial separator line between the two coil sides -----
553     plot([R_yoke_in*cos(slot_c), R_bore*cos(slot_c)], ...
554          [R_yoke_in*sin(slot_c), R_bore*sin(slot_c)], ...
555          'k-', 'LineWidth',0.6);
556
557     % --- Direction symbols and phase labels -----
558     rm = (R_yoke_in + R_bore)/2;
559     thR = slot_c - slot_half/2; % centre of right half (Layer 1)
560     thL = slot_c + slot_half/2; % centre of left half (Layer 2)
561     if dir_L1(k) > 0, s1 = '\bullet'; else, s1 = '\times'; end
562     if dir_L2(k) > 0, s2 = '\bullet'; else, s2 = '\times'; end
563     sgn1 = '+'; if dir_L1(k) < 0, sgn1 = char(8722); end
564     sgn2 = '+'; if dir_L2(k) < 0, sgn2 = char(8722); end
565
566     text(rm*cos(thR), rm*sin(thR), ...
567          {s1, [phase_letter{ph_L1(k)} sgn1]}, ...
568          'HorizontalAlignment','center','VerticalAlignment','middle','FontSize',7);
569     text(rm*cos(thL), rm*sin(thL), ...
570          {s2, [phase_letter{ph_L2(k)} sgn2]}, ...
571          'HorizontalAlignment','center','VerticalAlignment','middle','FontSize',7);
572
573     % --- Slot number -----
574     text(1.10*cos(slot_c), 1.10*sin(slot_c), num2str(k), ...
575          'HorizontalAlignment','center','FontSize',9,'FontWeight','bold');
576 end

```

```
577 % --- Rotor (visual reference) -----
578 fill(R_rotor*cos(th_full), R_rotor*sin(th_full), [0.95 0.90 0.75], ...
579      'EdgeColor','k','LineWidth',0.6);
580 text(0, 0, 'Rotor', 'HorizontalAlignment','center','FontSize',8, ...
581      'Color',[0.4 0.4 0.4]);
582
583
584 xlim([-1.25 1.25]); ylim([-1.25 1.25]);
585 title(ttl, 'FontSize', 11);
586 legend([patch(NaN,NaN,cA), patch(NaN,NaN,cB), patch(NaN,NaN,cC)], ...
587        'Phase A','Phase B','Phase C', ...
588        'Location','southoutside','Orientation','horizontal');
589 end
```